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Laor

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(54) **SMOOTH SKIPPING AND ROLO-SKIP ELECTRICAL MOTORS**

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(52) **U.S. Cl.**
CPC **H02K 41/065** (2013.01); **H02K 2213/03** (2013.01)

(58) **Field of Classification Search**
CPC H02K 41/00; H02K 41/06; H02K 41/065;
H02K 37/00; H02K 37/18; H02K 7/14
See application file for complete search history.

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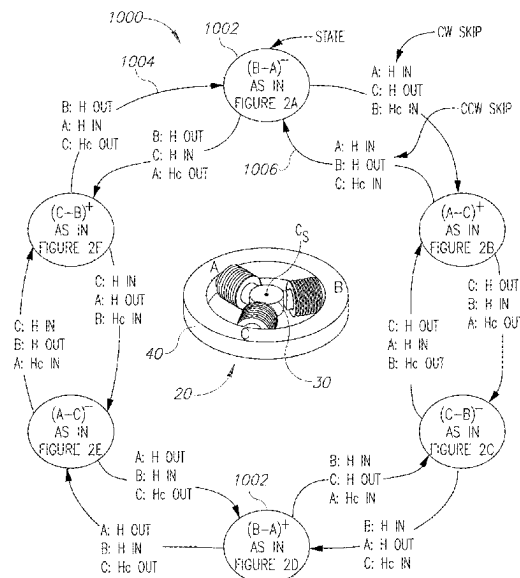
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(57) **ABSTRACT**

A skipper motor comprising: a stator and a rotor; a plurality of N greater than two electromagnets mounted to the stator or to the rotor, each of the electromagnets comprising a core having at least one contact surface, wherein the at least one contact surface of any of the cores is rotatable about a same first axis of rotation with a same first radius of rotation into substantial congruence with at least a portion of the at least one contact surface of any other of the electromagnet cores; and a coupling of the rotor to the stator configured to enable rotation of the rotor and contact of the stator and rotor along at least one contact surface during operation of the motor.

18 Claims, 16 Drawing Sheets



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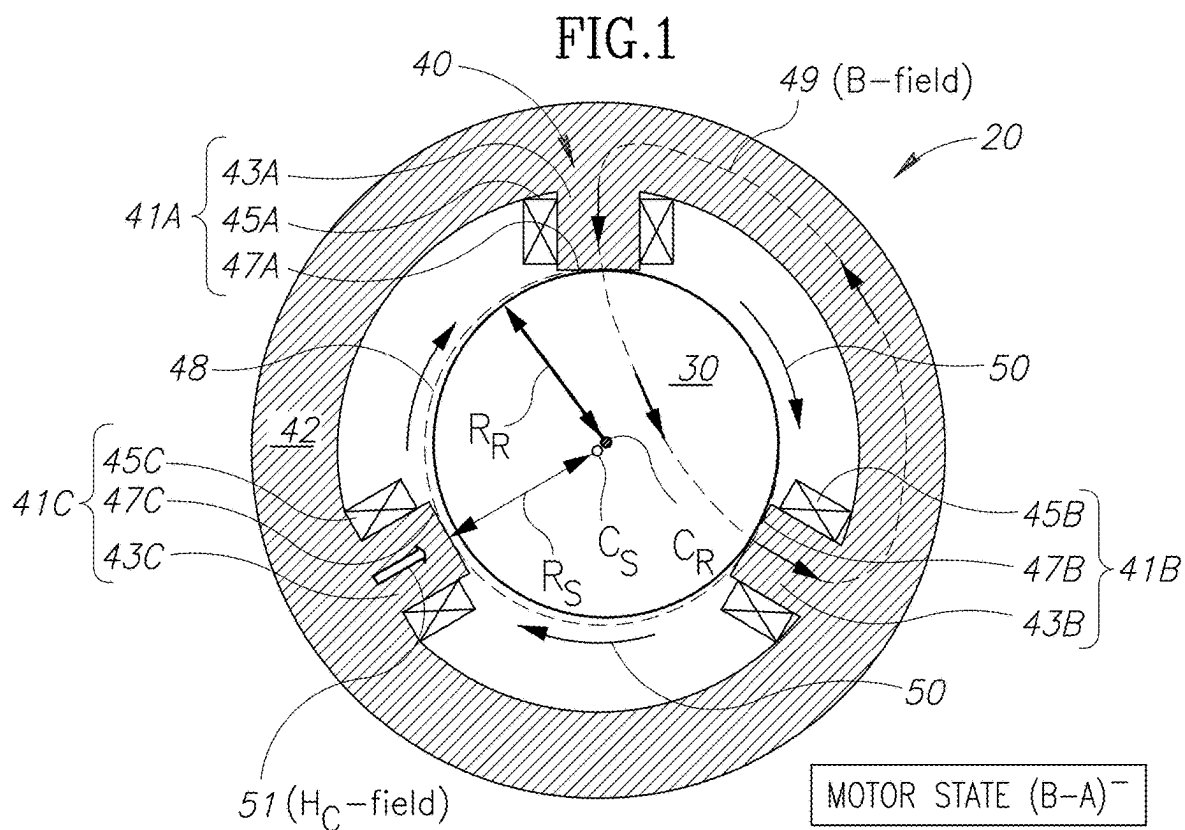
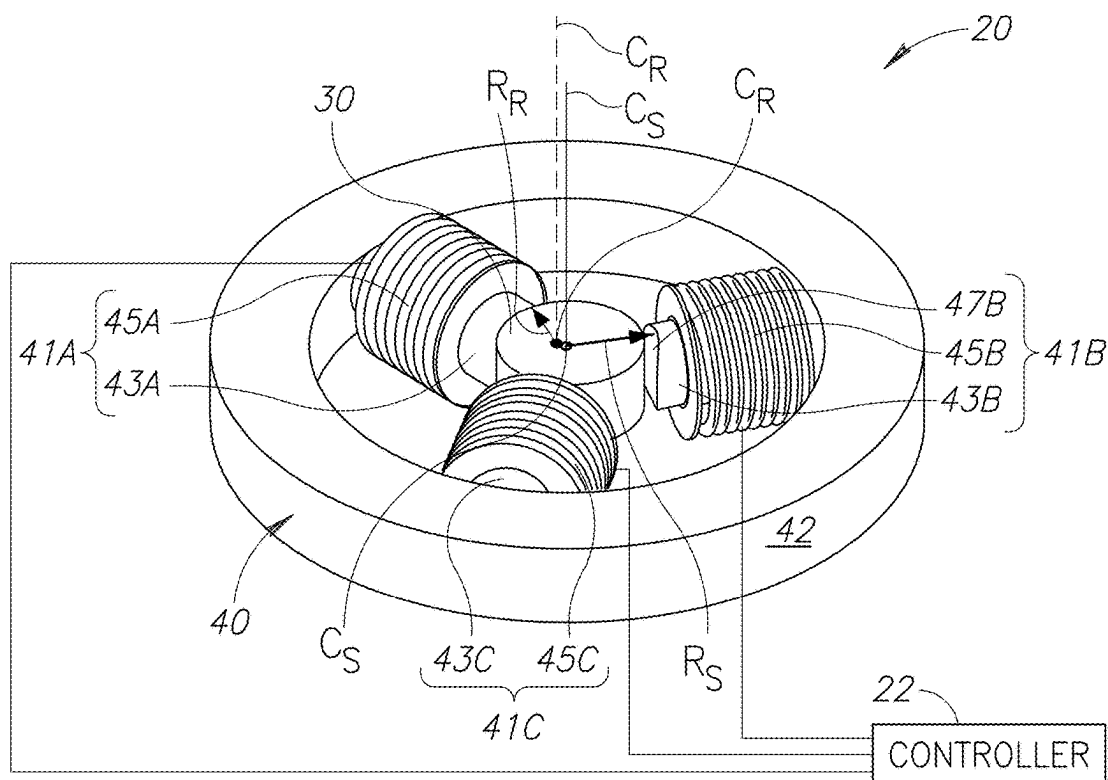
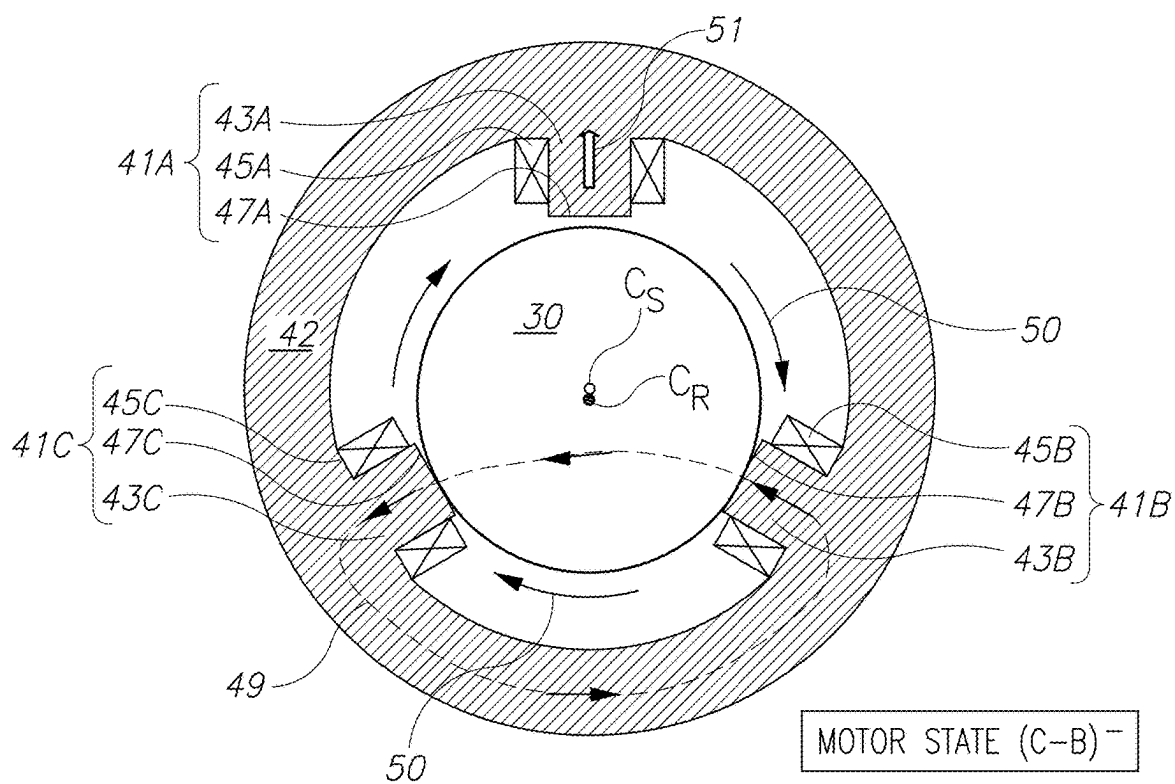
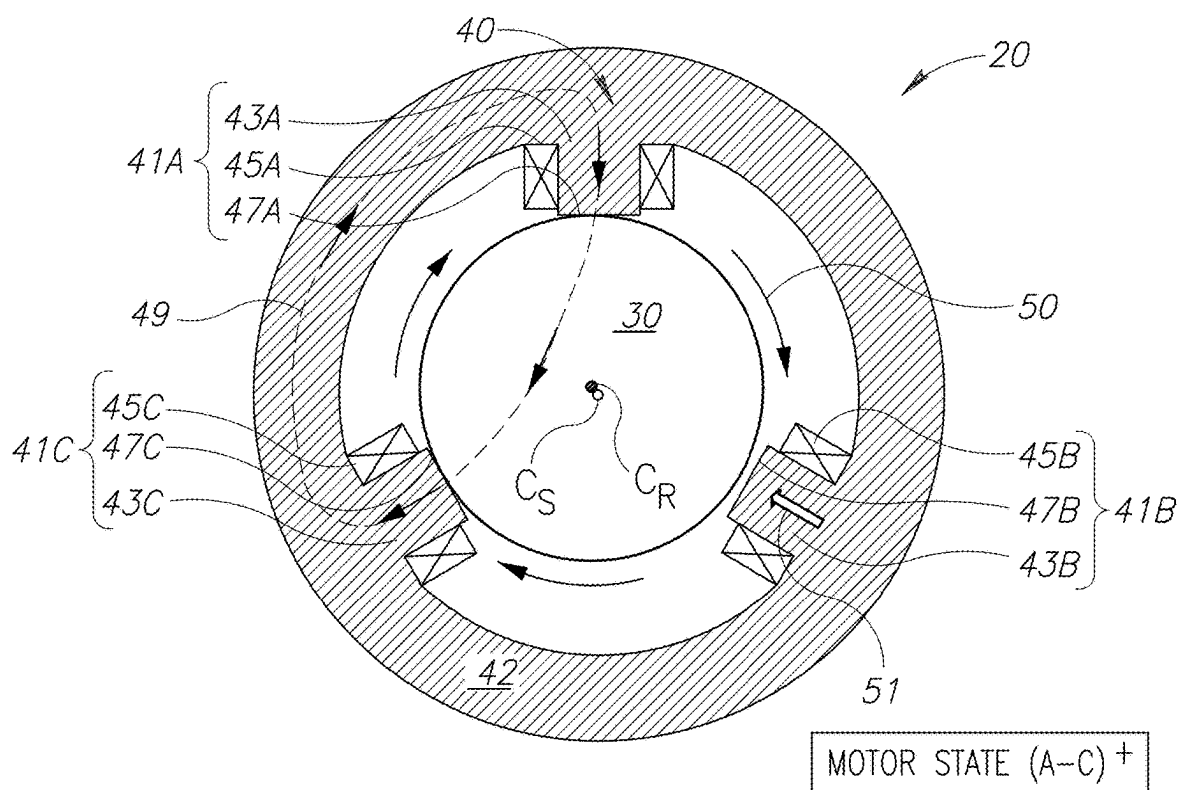
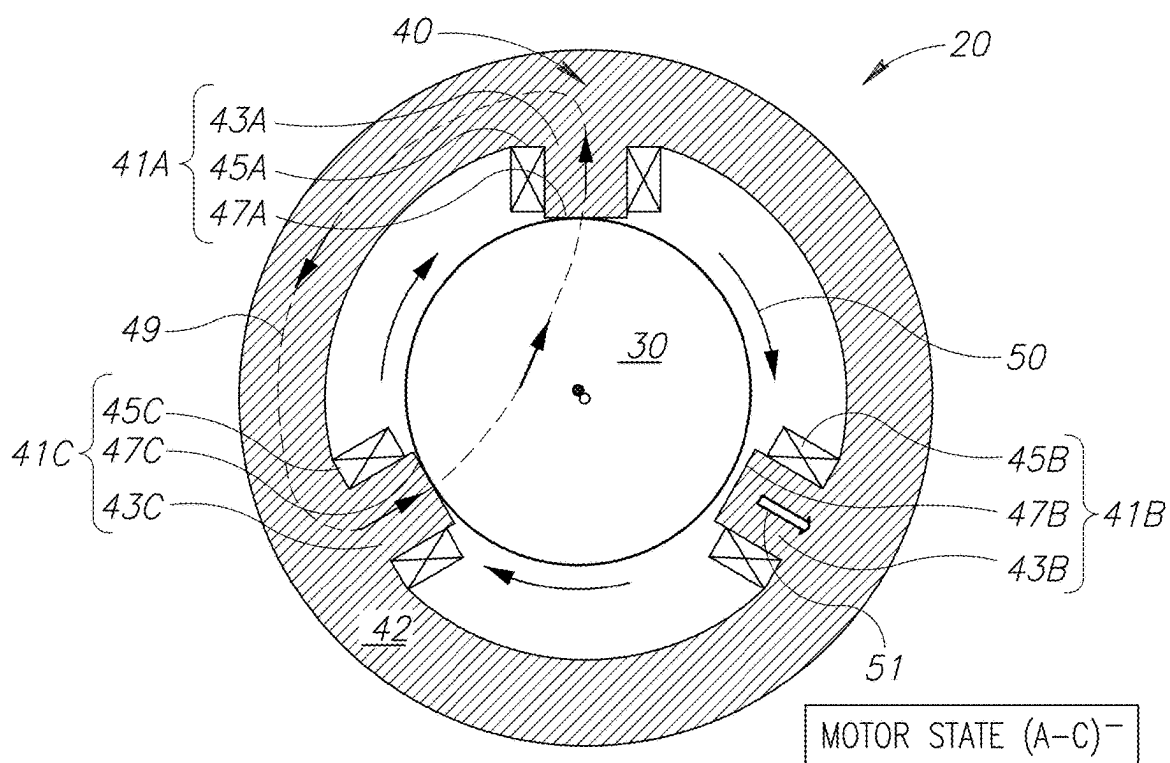
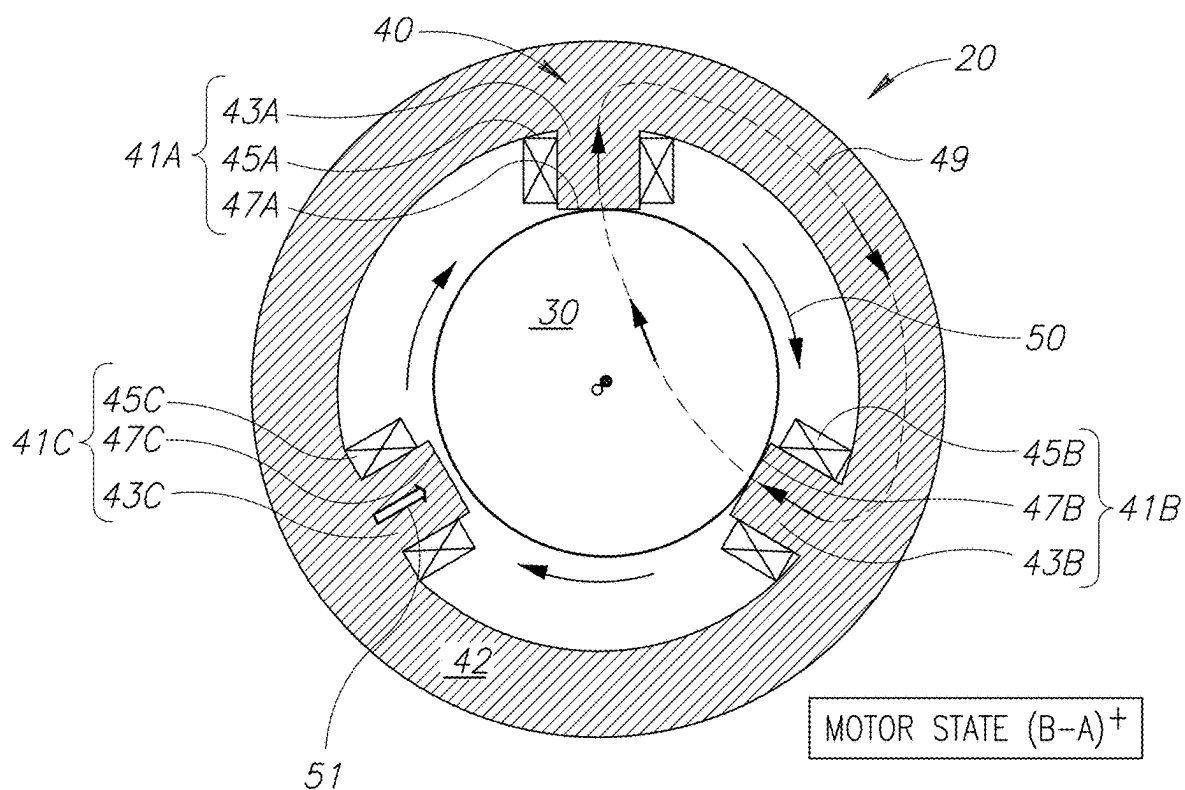


FIG. 2A





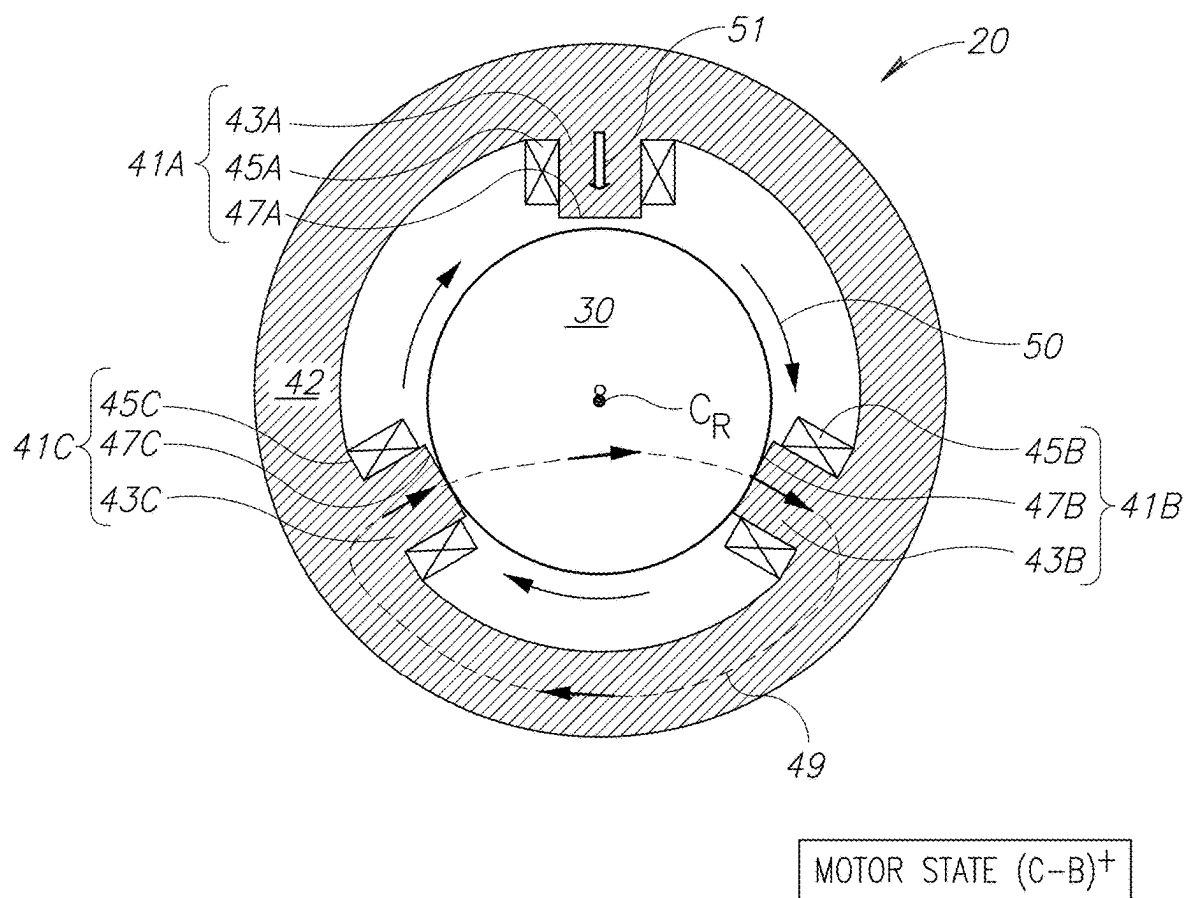


FIG.2F

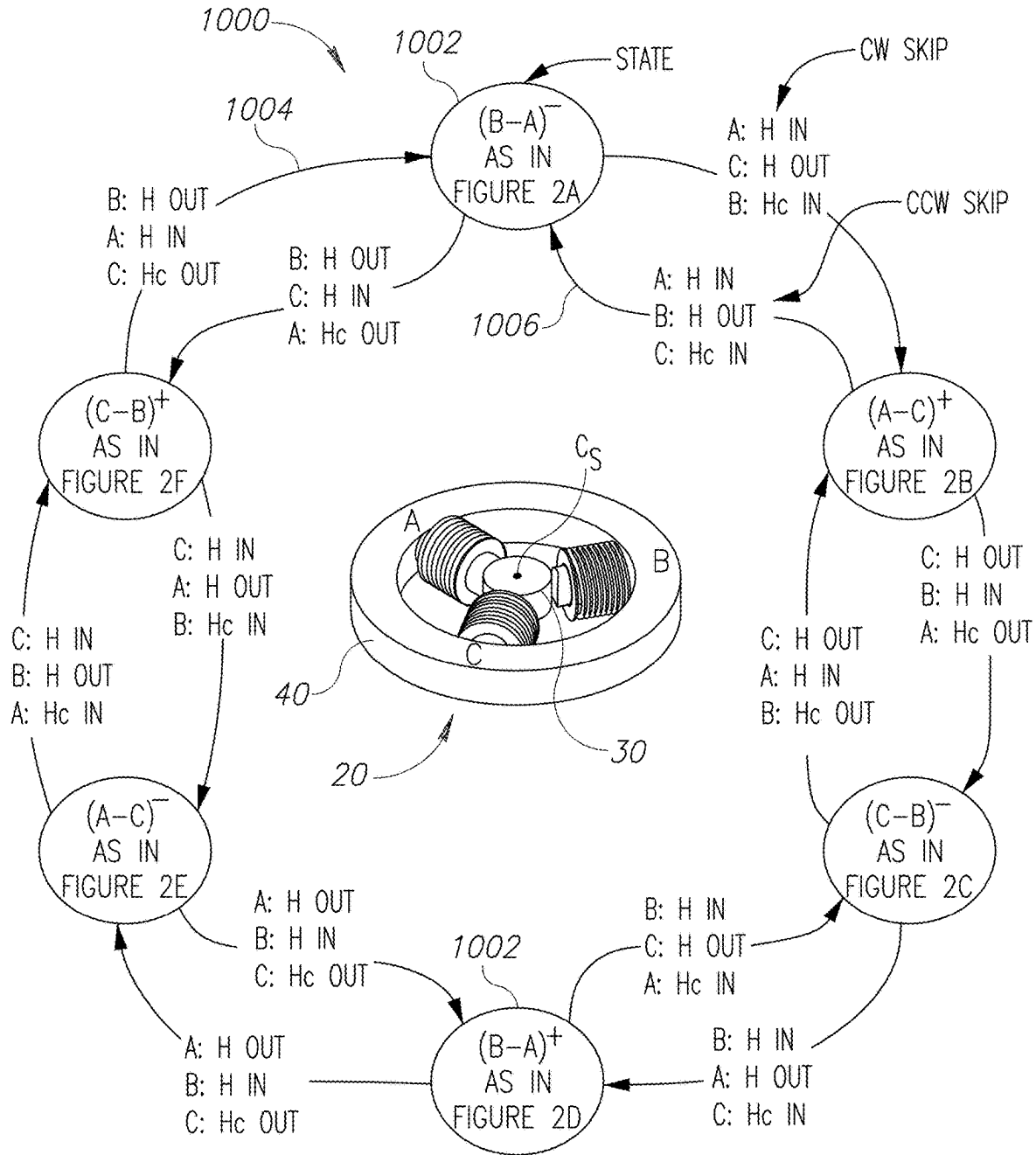


FIG. 2G

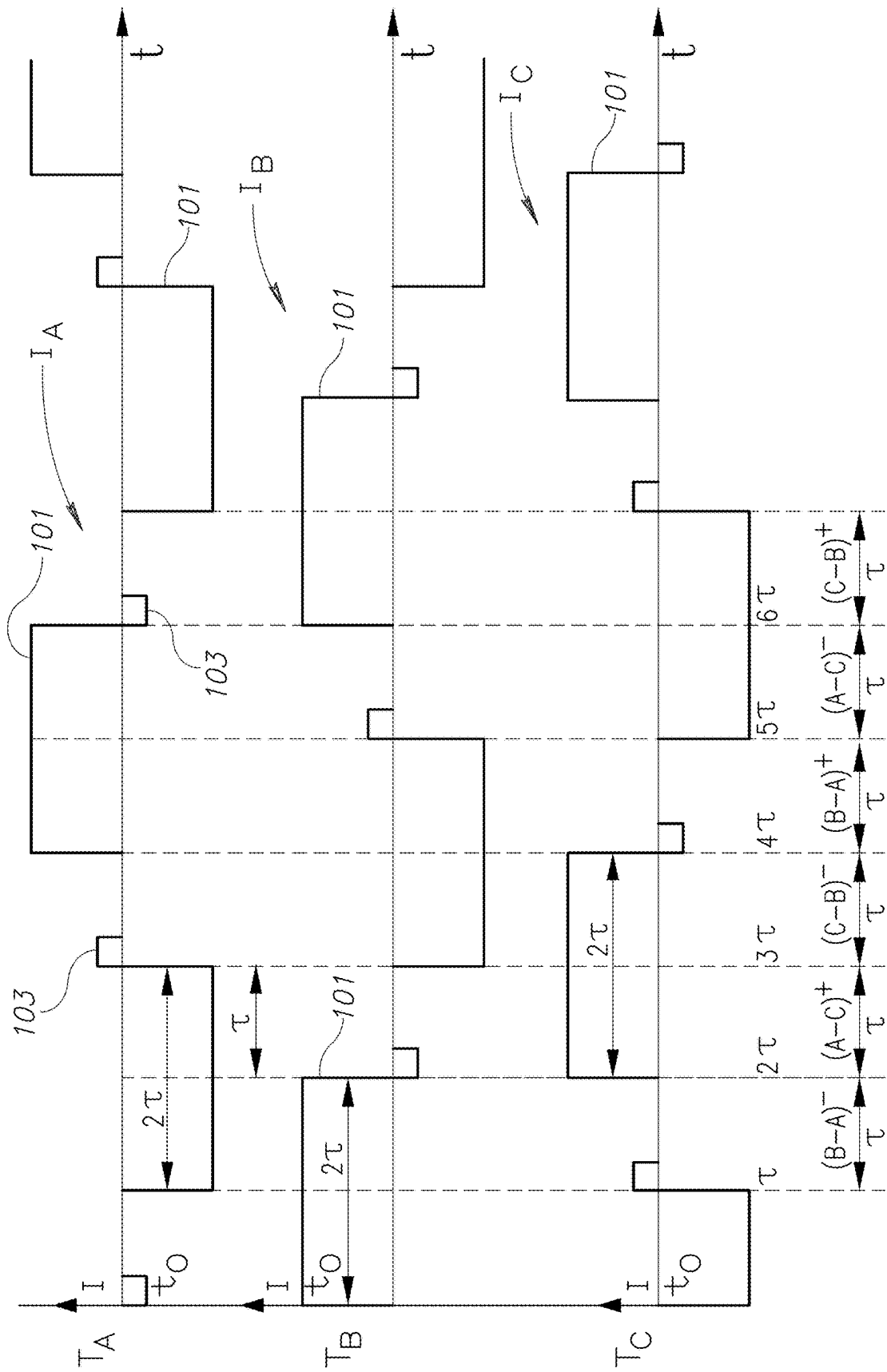


FIG.3A

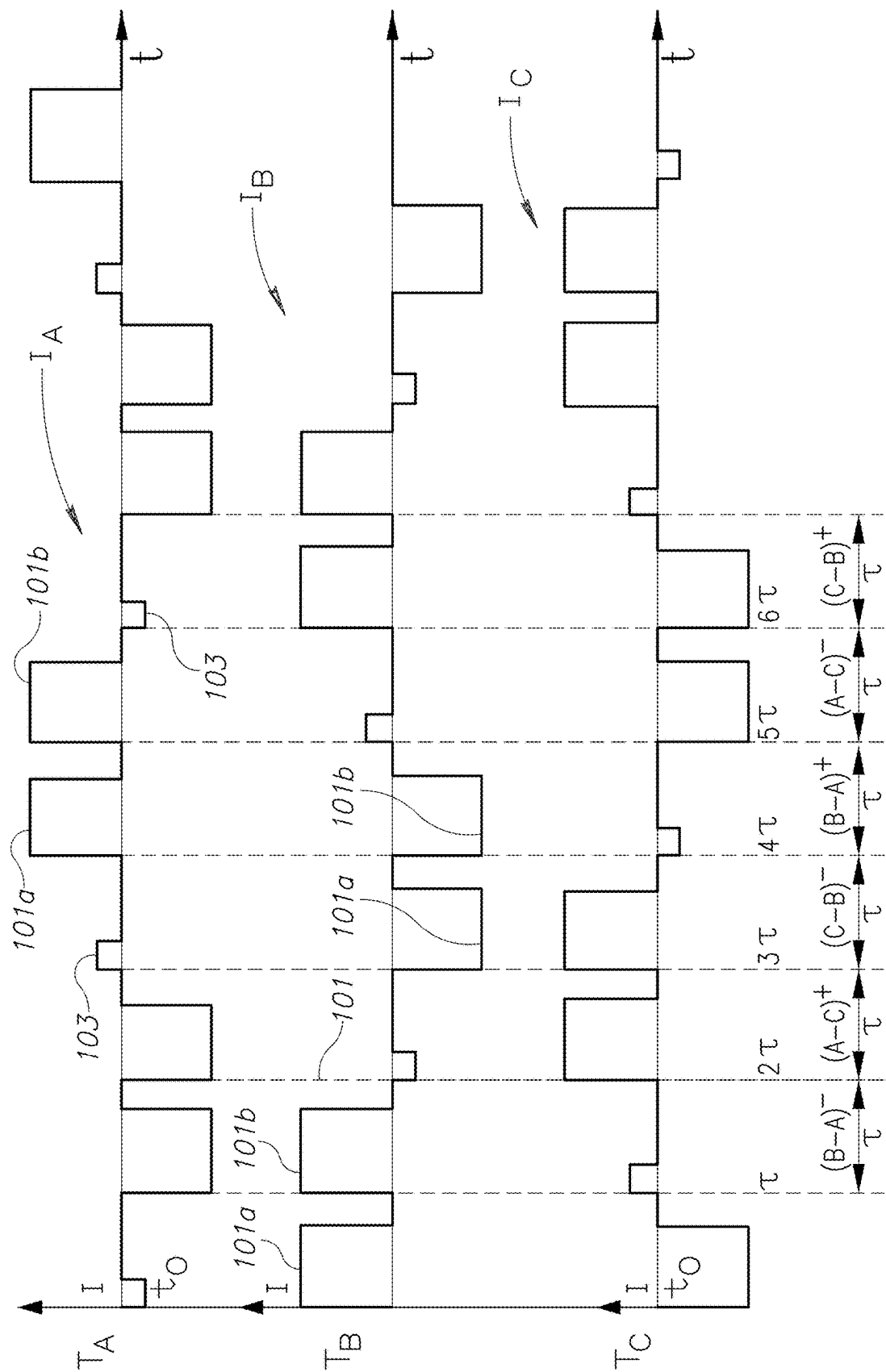


FIG.3B

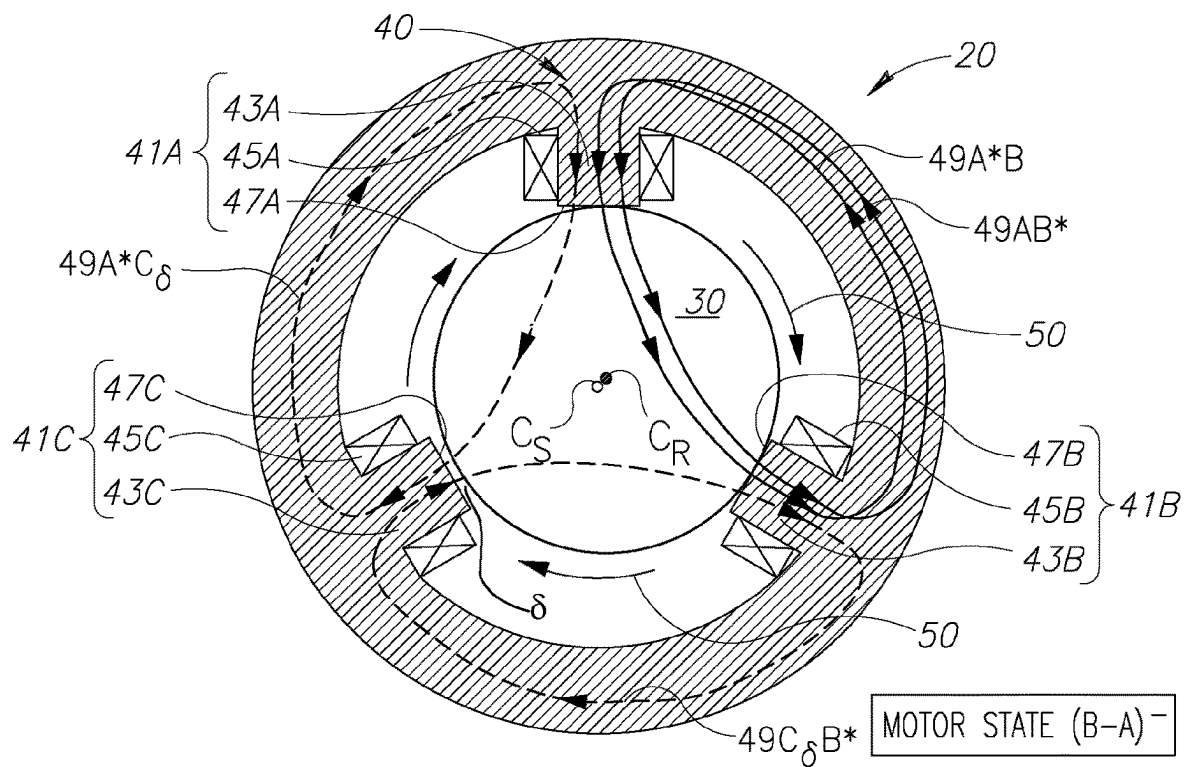


FIG.3C

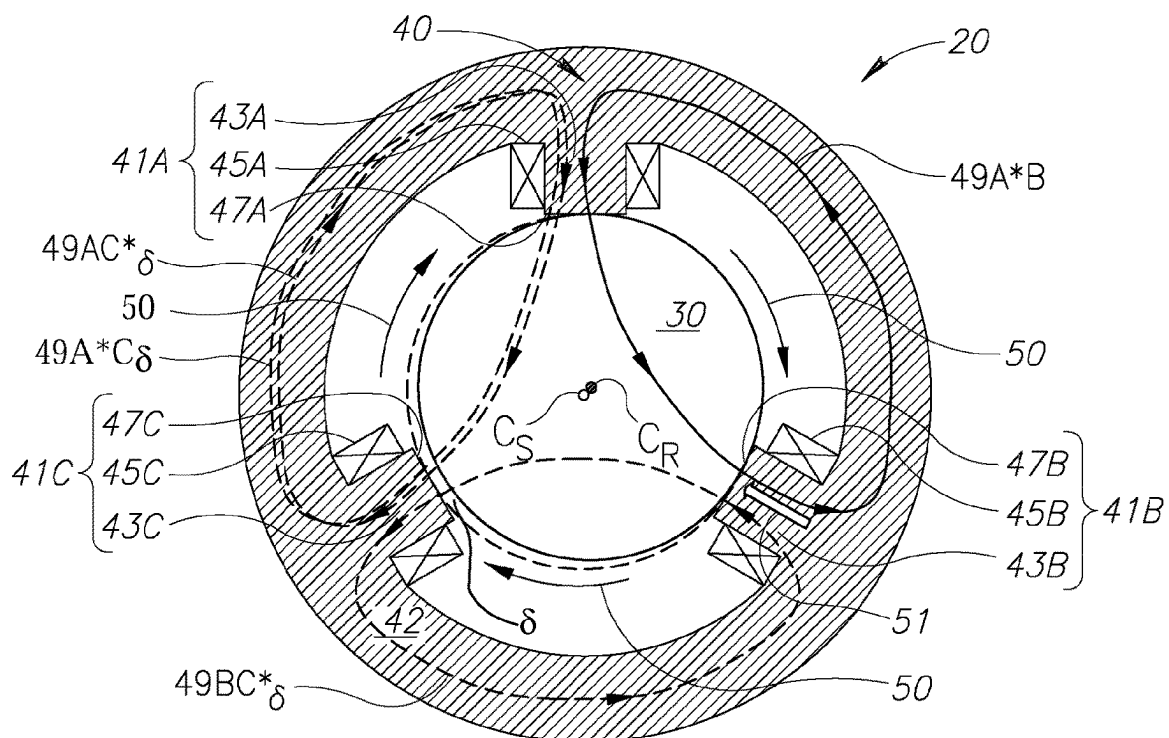
MOTOR SKIPPING FROM STATE (B-A)⁻ TO STATE (A-C)⁺

FIG.3D

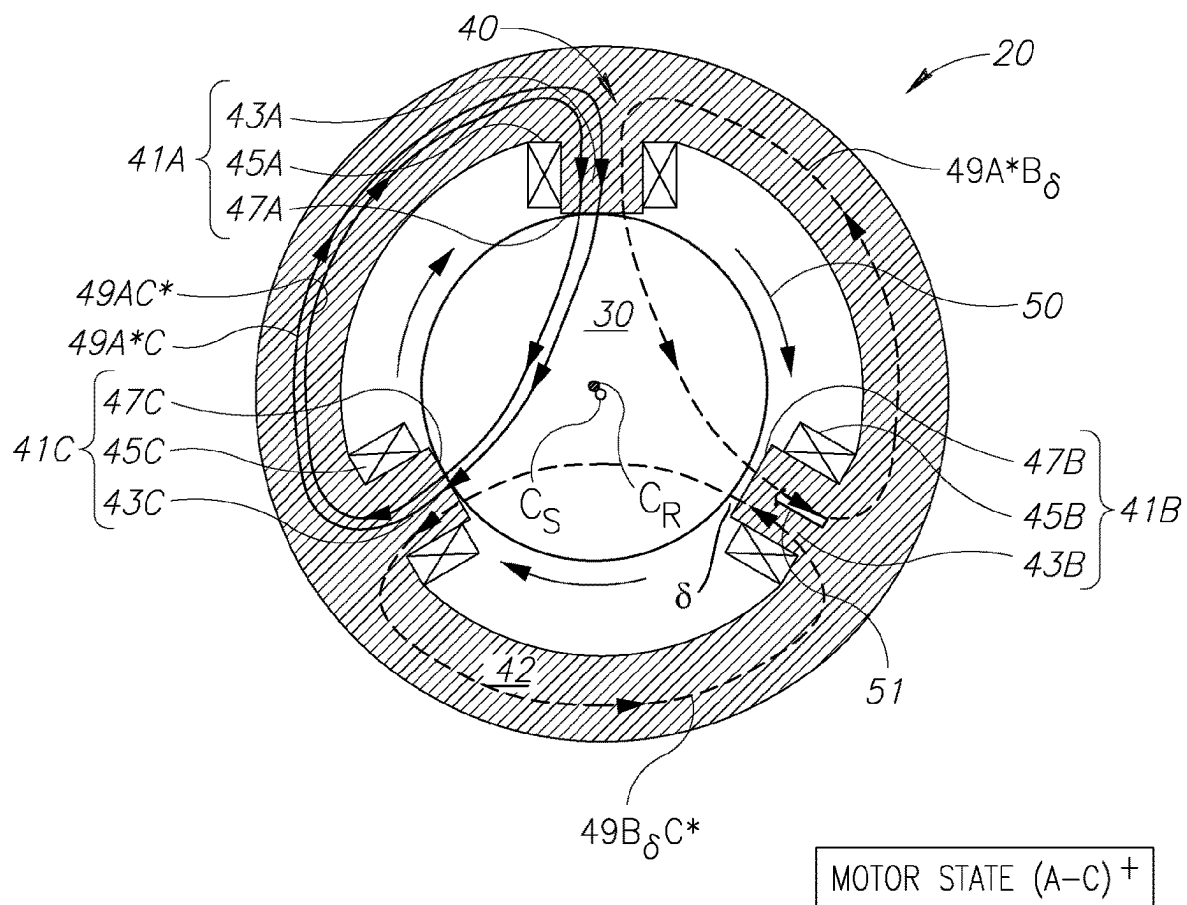


FIG. 3E

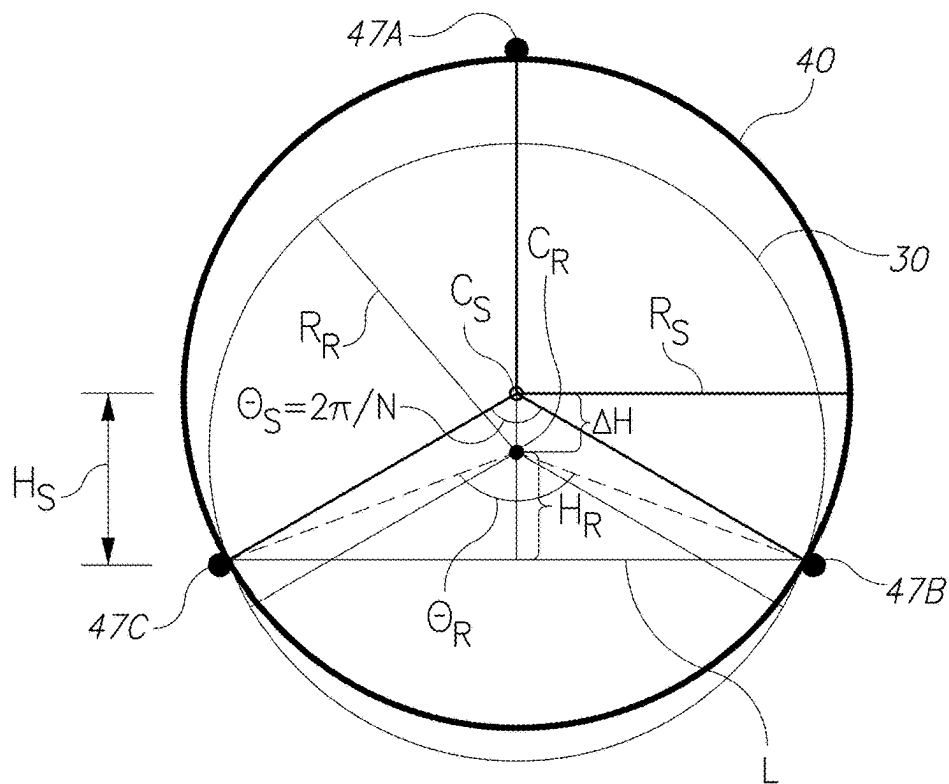


FIG. 4A

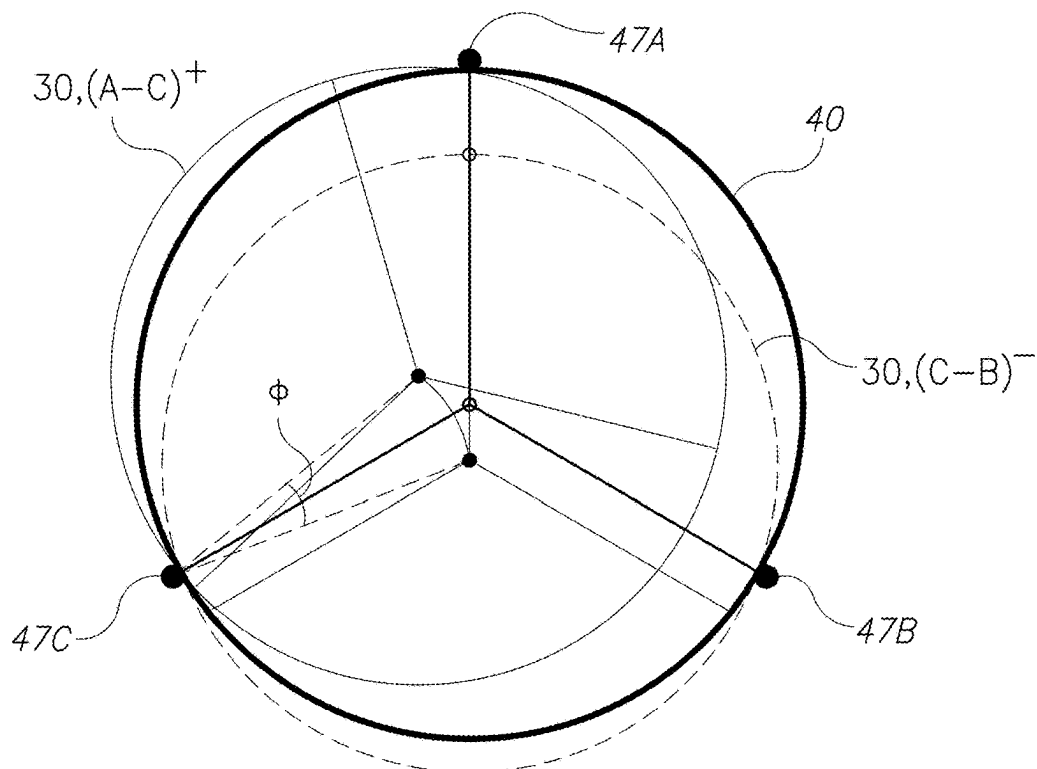


FIG. 4B

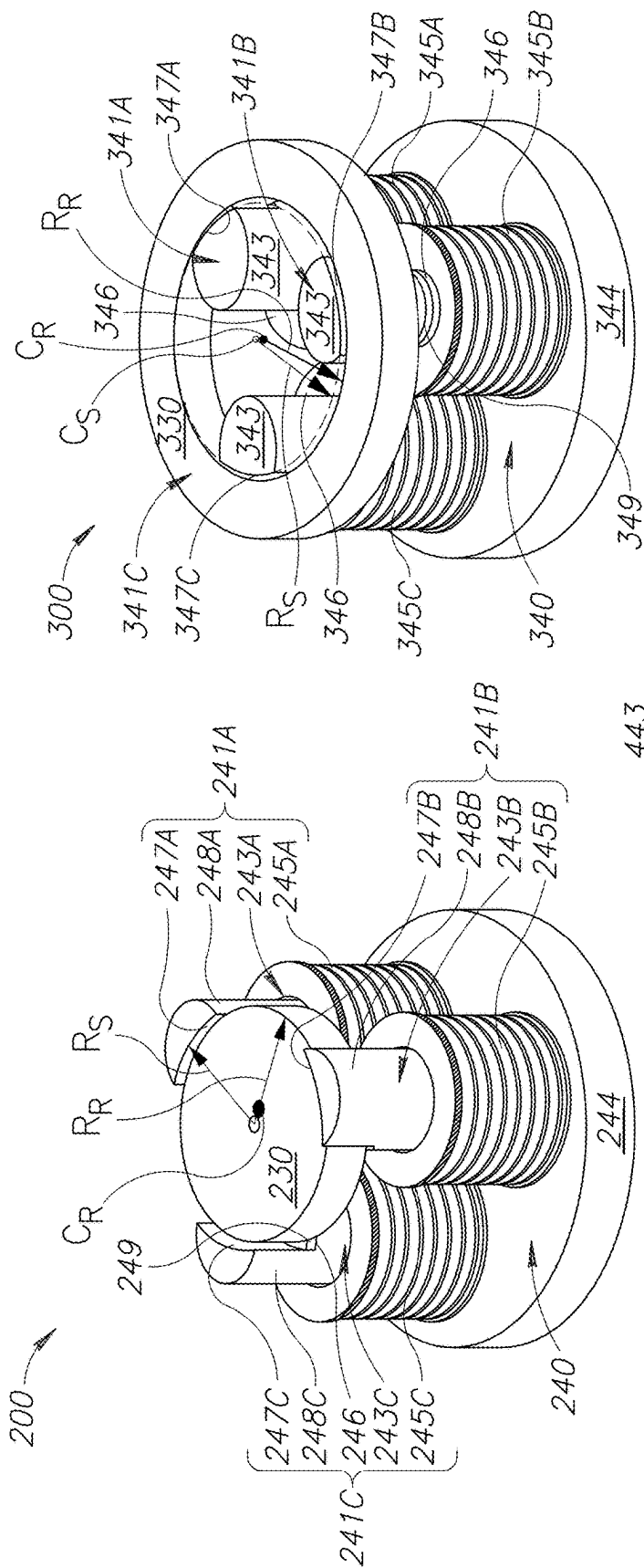


FIG. 5A

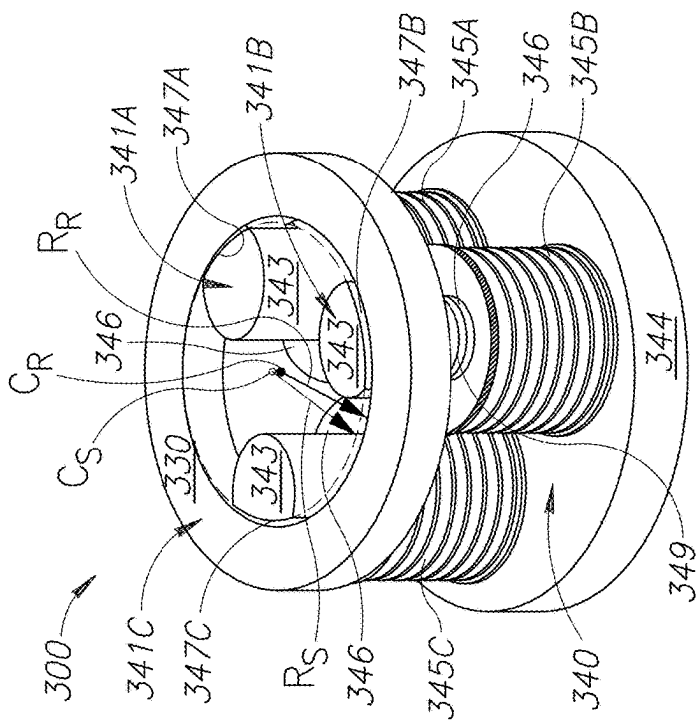


FIG. 5B

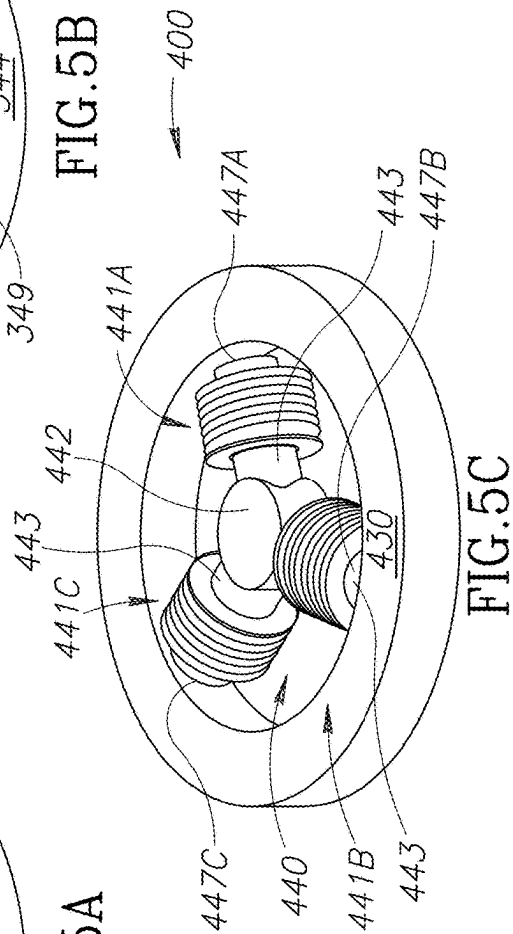


FIG. 5C

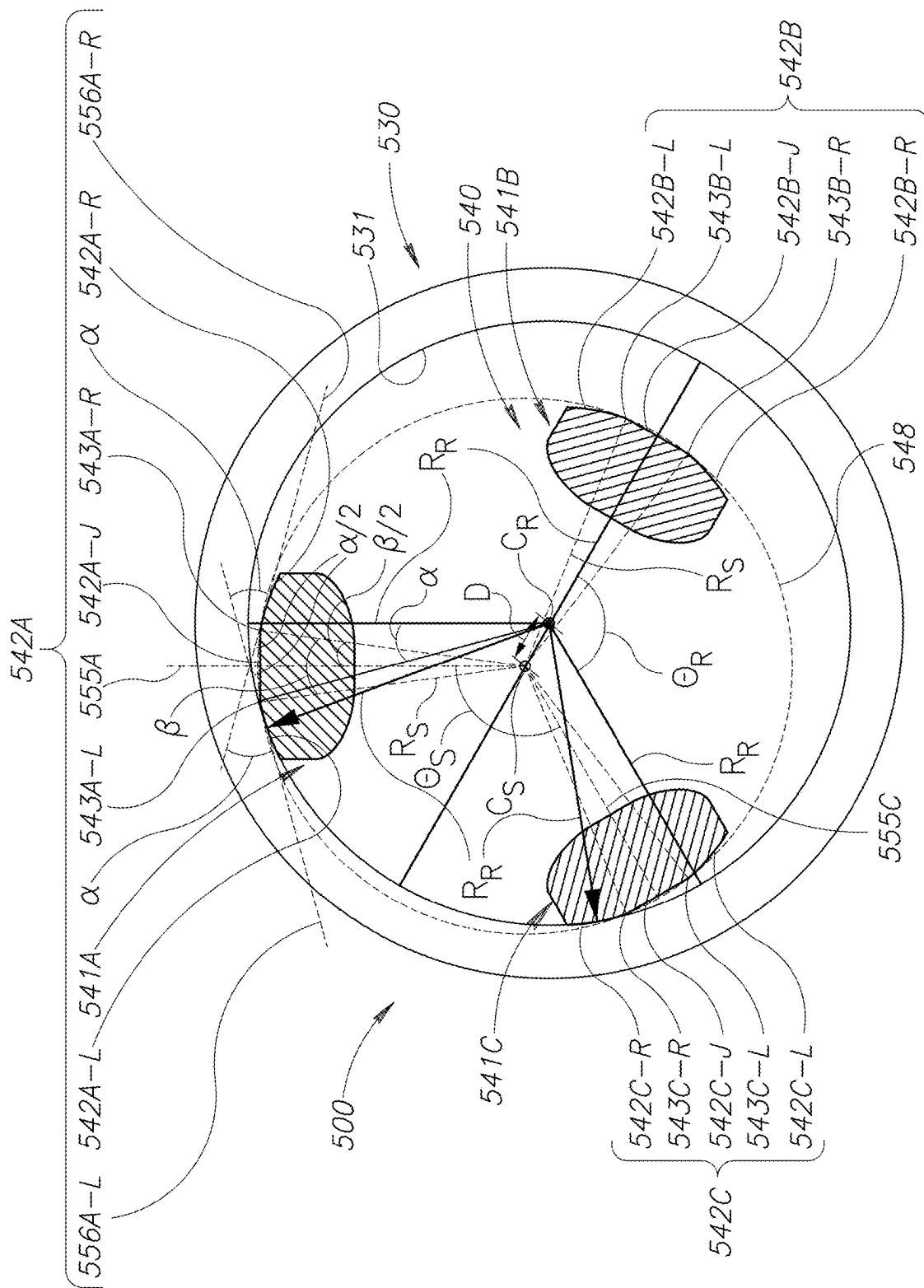


FIG. 6A

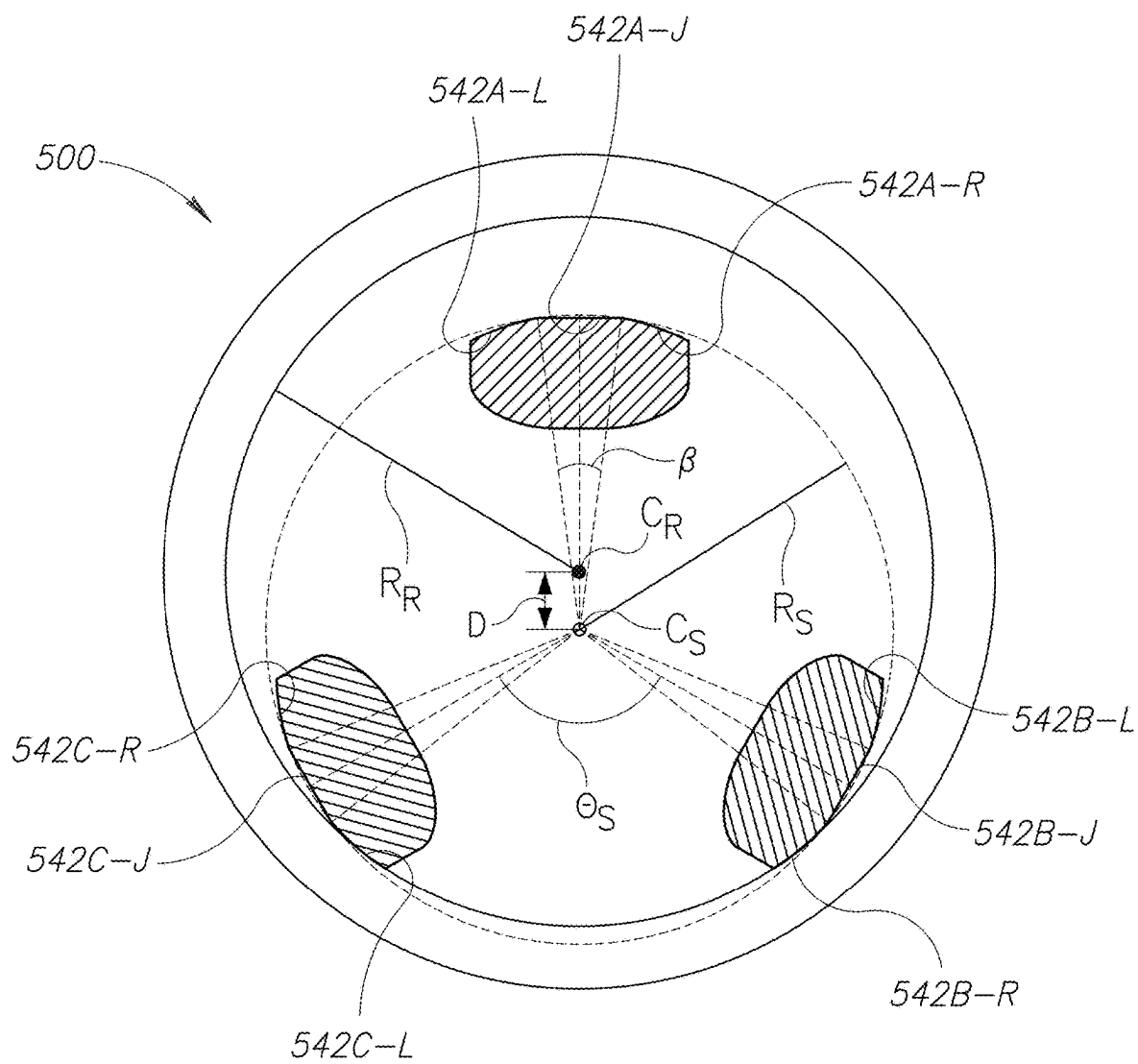


FIG. 6B

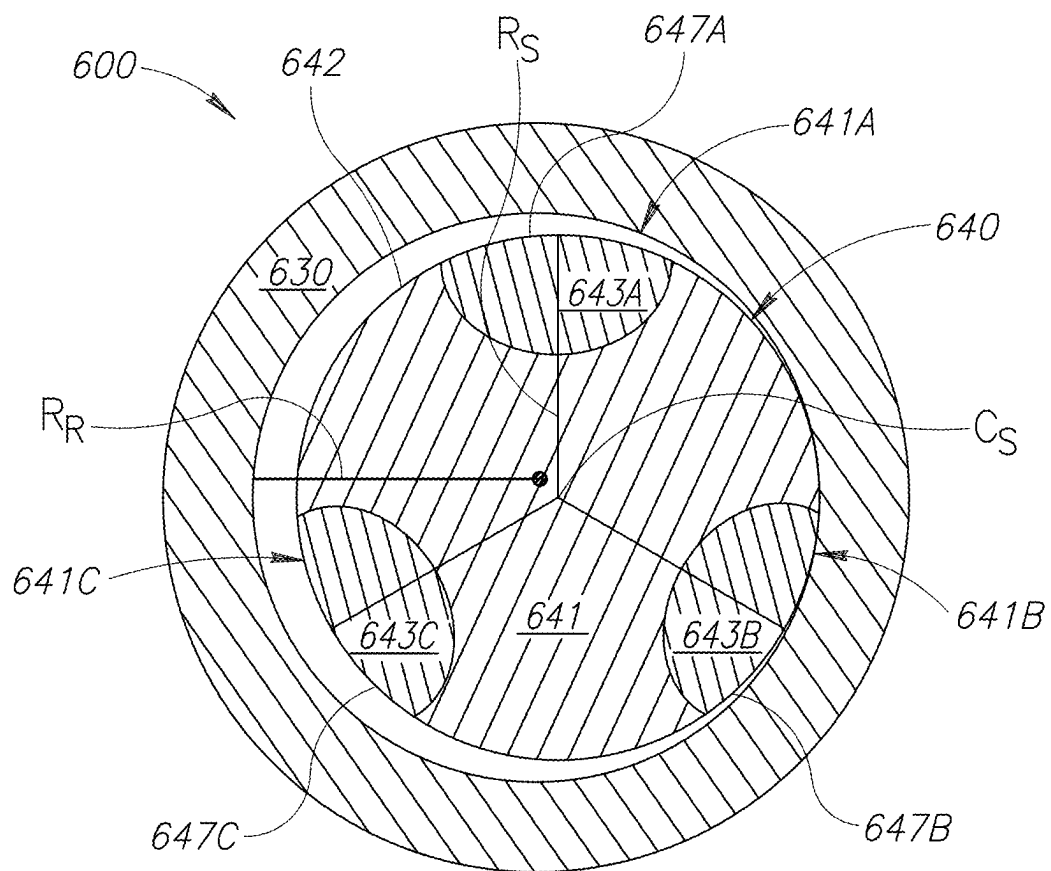


FIG. 7

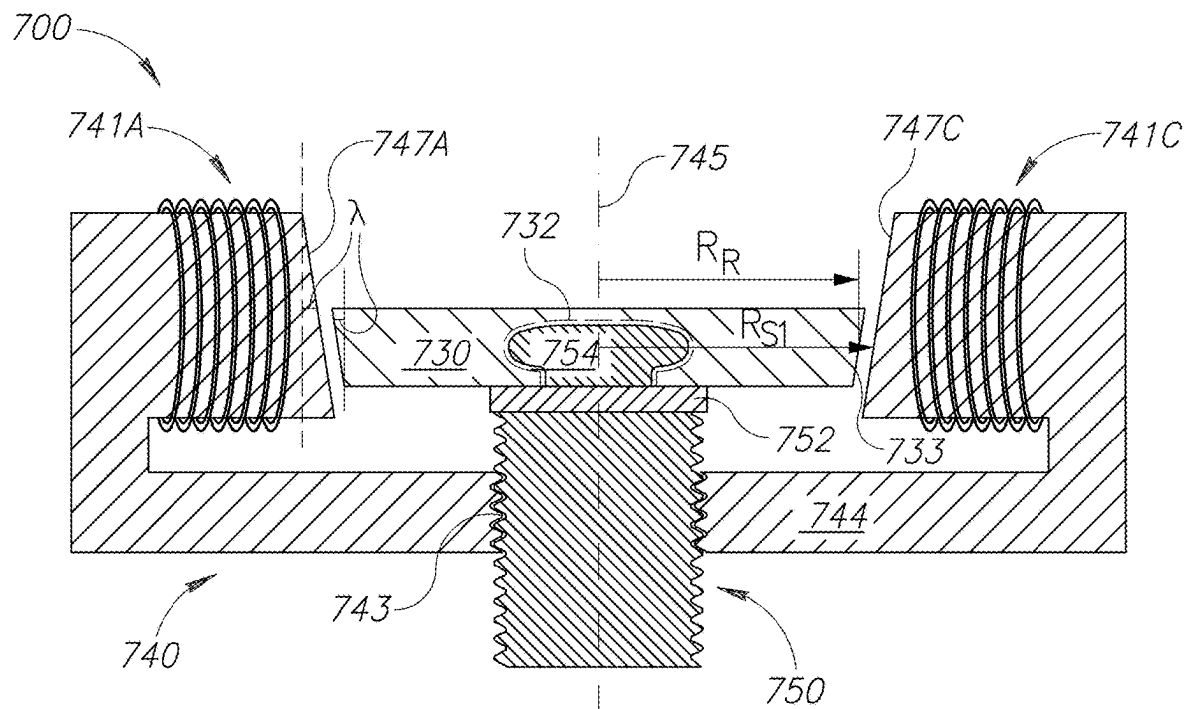


FIG. 8A

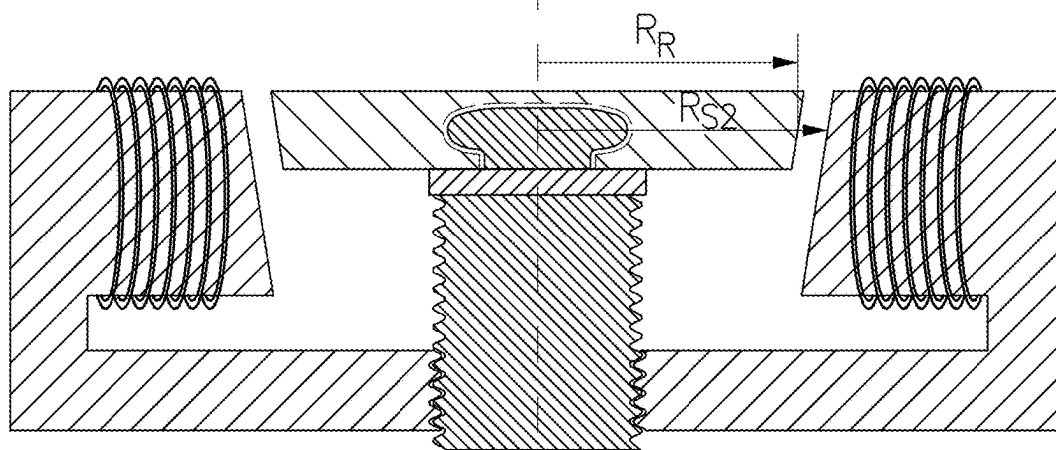


FIG. 8B

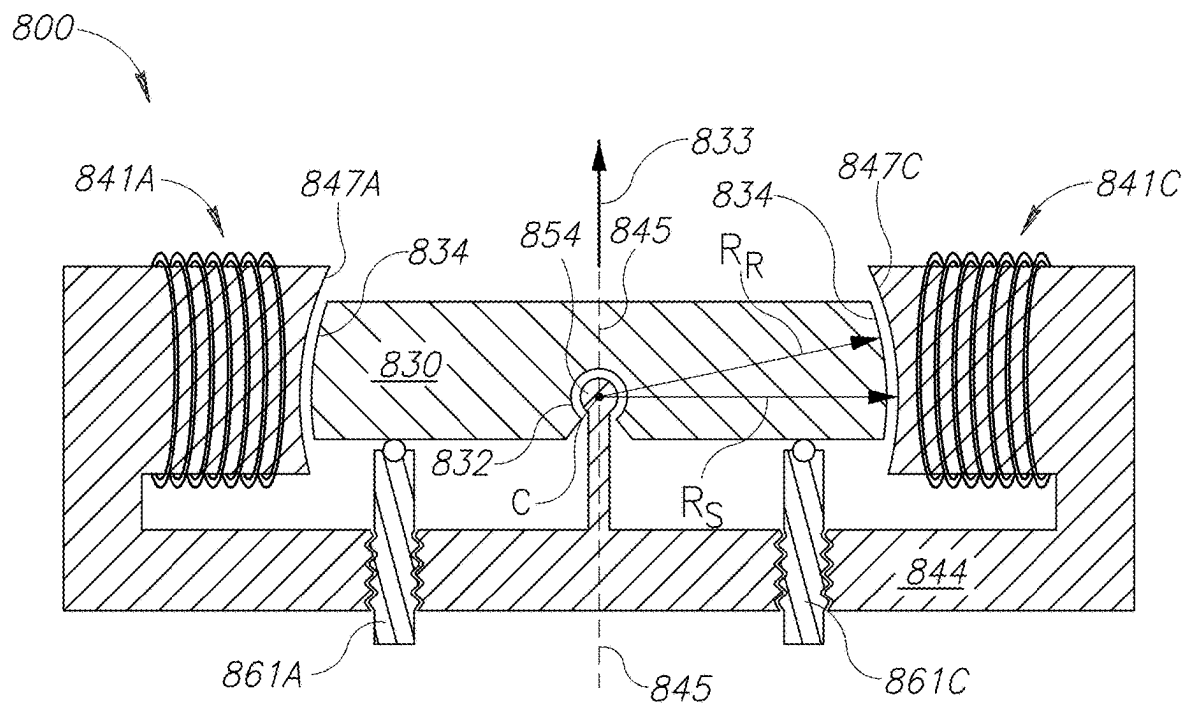


FIG. 9A

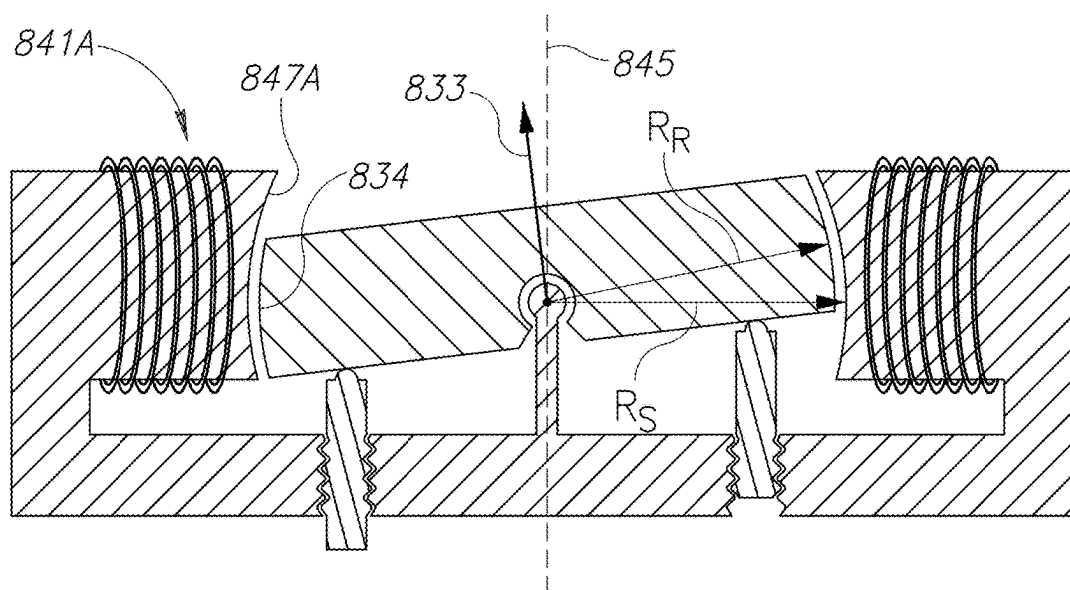


FIG. 9B

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SMOOTH SKIPPING AND ROLO-SKIP ELECTRICAL MOTORS

RELATED APPLICATIONS

The present application claims benefit under 35 U.S.C. 119(e) of U.S. Provisional Application 63/515,134 filed Jul. 24, 2023, the disclosure of which is incorporated herein by reference in their entirety.

FIELD

Embodiments of the specification relate to electric motors.

BACKGROUND

Electric motors are ubiquitous and come in a plethora of various forms adapted to perform an ever-increasing gamut of applications. From the humble beginnings of Faraday's 1821 demonstration of a current carrying wire rotating around a permanent magnet, electric motors have developed in performance, configuration, and range of power and complexity. Today they may be found by way of limited examples in such disparate applications and environments as children's toys, perambulating, swimming, and flying robots and micro-robots, flying drones, submarines, ground vehicles, and spaceships.

SUMMARY

An aspect of an embodiment of the disclosure relates to providing a versatile electric stepper motor, also referred to as a "skipping motor", in which a rotor of the motor nutates as it rotates clockwise or counterclockwise to execute a skipping-like motion between steps, also referred to as "skips", of the motor. At the end of each skip a magnitude of the nutation is substantially maximum relative to a center of the stator, and the rotor contacts the stator along a set of at least one contact surface of a plurality of contact surfaces of the stator and at least one corresponding contact surface of the rotor. A configuration of rotor and stator contact surfaces that are in contact at an end of a given skip and associated configuration of the magnetization of the contact surfaces may be referred to as a motor state. Immediately sequential motor states are different from each other and rotated relative to each other by a skip angle.

In an embodiment, each contact surface of the stator comprises a surface of an electromagnet of a plurality of, "N", three or more electromagnets, which may be configured to exhibit an N-fold rotational symmetry characterized by a symmetry angle of rotation $2\pi/N$ about the center of the stator. Contact surfaces of the rotor are surfaces of magnetizable regions, also referred to as cores or poles, of the rotor. Sequentially magnetizing and de-magnetizing each electromagnet of the N electromagnets in accordance with an ON-OFF activation sequence that is shifted in time with respect to the ON-OFF activation sequences of the other electromagnets causes the rotor to sequentially skip through 2N skips of the skipping motor before returning to a same motor state if the motor comprises an odd number of electromagnets and rotate clockwise or counterclockwise. If the motor comprises an even number of electromagnets the motor skips through N skips of the motor in rotating clockwise or counterclockwise. At each motor state the rotor is attracted by magnetic fields of the electromagnets so that at least one contact surface of a magnetized stator electro-

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magnet contacts a corresponding contact surface of the rotor and magnetizes the magnetizable region of the rotor associated with the rotor contact surface.

A given motor state of a skipping motor may be distinguished by a label that designates the particular contact surfaces that are in contact when the skipping motor is in the given state. In addition, a label may designate a direction of a magnetic field that is generated to bring the contact surfaces into contact.

In an embodiment, the ON-OFF activation sequences of the electromagnets are configured so that at each motor state contact between the rotor and stator provides a relatively low reluctance magnetic field path for a magnetic field generated by magnetization of at least two magnetized electromagnets of the N electromagnets.

In an embodiment, a number of skips in one complete rotation of the skipping motor rotor and a corresponding skip angle of the rotor per skip may be a function of a ratio, hereinafter also referred to as a skipping ratio, between a radius of a circularly cylindrical surface on which rotor contact surfaces lie and a radius of a circularly cylindrical surface of the stator on which stator contact surfaces lie. Optionally, the skipping motor is controllable to change a skipping ratio of the skipping motor and operate at each of a plurality of different selectable skipping ratios. Optionally the surfaces on which the stator and/or rotor contact surface are respectively located are circularly cylindrical or spherical surfaces. The spherical surfaces enable the skipping motor to operate with an axis of rotation of the rotor angled at different desired tilt angles with respect to an axis of symmetry of the stator.

In an embodiment each electromagnet of a skipping motor has two contact surfaces and in skipping from a first motor state to a second motor state first one and then the other of the two contact surfaces contact the rotor.

Whereas the electromagnets of a skipping motor are described above as comprised in a stator of a skipping motor, in an embodiment the electromagnets and their associated contact surfaces may be comprised in a rotor of a skipping motor and corresponding contact surfaces associated with magnetizable regions may be comprised in a stator of the motor.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

BRIEF DESCRIPTION OF FIGURES

Non-limiting examples of embodiments of the invention are described below with reference to figures attached hereto that are listed following this paragraph. Identical features that appear in more than one figure are generally labeled with a same label in all the figures in which they appear. A label labeling an icon representing a given feature of an embodiment of the invention in a figure may be used to reference the given feature. Dimensions of features shown in the figures are chosen for convenience and clarity of presentation and are not necessarily shown to scale.

FIG. 1 schematically shows a skipping motor comprising a rotor and a stator having electromagnets, in accordance with an embodiment of the disclosure;

FIGS. 2A-2F schematically show the motor shown in FIG. 1 with the electromagnets being excited to drive the rotor clockwise, in accordance with an embodiment of the disclosure;

FIG. 2G shows a state diagram that schematically illustrates both clockwise and counterclockwise rotation of the skipping motor shown in FIGS. 1-2F, in accordance with an embodiment of the disclosure;

FIGS. 3A and 3B show graphs of activation sequences as a function of time for each of the electromagnets of the skipping motor shown in FIGS. 1-2F that generate clockwise rotation of the skipping motor, in accordance with an embodiment of the disclosure;

FIGS. 3C-3E schematically illustrate another method, optionally referred to as a Smooth-Skip method, of activating electromagnets in the skipping motor shown in FIG. 1 in accordance with an embodiment of the disclosure.

FIGS. 4A and 4B schematically illustrate parameters of the skipping motor shown in FIGS. 1-2F that determine a skip angle of the motor in accordance with an embodiment of the disclosure;

FIGS. 5A-5C schematically show different configurations of a skipping motor in accordance with an embodiment of the disclosure;

FIGS. 6A and 6B schematically show a skipping motor having two contact surfaces for each electromagnetic comprised in the motor, in accordance with an embodiment of the disclosure;

FIG. 7 schematically shows a skipping motor configured to provide a relatively smooth, rolling skip between motor states of the motor, in accordance with an embodiment of the disclosure;

FIGS. 8A and 8B schematically shows a skipping motor for which a skipping ratio is changeable, in accordance with an embodiment of the disclosure; and

FIGS. 9A and 9B schematically show a skipping motor for which an angle between an axis of rotation of a rotor in the skipping motor may be changed relative to an axis of the stator of the skipping motor, in accordance with an embodiment of the disclosure.

DETAILED DESCRIPTION

FIG. 1 schematically shows a skipping motor 20 comprising a circularly cylindrical ferromagnetic rotor 30 located inside a stator 40 having, optionally three, electromagnets 41A, 41B, and 41C, supported by a cylindrical shell shaped support ring 42, in accordance with an embodiment of the disclosure. Rotor 30 has a radius of rotation RR about an axis of rotation represented by a dashed line labeled CR that passes through a central point of the rotor indicated by a solid circle also labeled CR. Central point CR may be referred to as a center CR and axis of rotation CR may be referred to and represented by center CR. Electromagnets 41A, 41B, and 41C comprise ferromagnetic cores 43A, 43B, 43C respectively that extend from ferromagnetic support ring 42. Activation coils 45A, 45B, 45C of electrically conductive wire respectively surround cores 43A, 43B, 43C. Cores 43A, 43B, 43C have respective end surfaces 47A, 47B, 47C, of which only end surface 47B is shown in the perspective of FIG. 1. End surfaces 47A, 47B, 47C are substantially tangent to a same cylindrical surface having radius RS about axis of rotation represented by a solid line labeled CS that passes through a central point of stator 40 indicated by a circle also labeled CS. Central point CS may be referred to as a center CS and the axis of rotation may be referred to and represented by center CS.

FIG. 2A discussed below schematically shows a plan view of skipping motor 20 in which the cylindrical surface having radius RS and center CS to which end surfaces 47A, 47B, 47C are tangent is indicated by a dashed circle 48 having radius RS and center CS. End surfaces 47A, 47B, and 47C are stator contact surfaces which rotor 30 contacts at different motor states during operation of skipping motor 20 as discussed below with reference to FIGS. 2A-2F. In an embodiment support ring 42 and cores 43A, 43B, 43C are integrally formed from a same ferromagnetic material. A controller 22 is configured to drive current through activation coils 45A, 45B, 45C to magnetize and demagnetize electromagnets 41A, 41B, and 41C and generate nutating, skipping, rotary motion of rotor 30, in accordance with an embodiment of the disclosure. Driving a current through an activation coil may be referred to as exciting or energizing the coil or electromagnet.

FIGS. 2A-2F schematically show plan views of skipping motor 20 at sequential motor states during operation of the motor in which controller 22 controls the motor to generate, optionally clockwise, rotary skipping motion of rotor 30, in accordance with an embodiment of the disclosure. Direction of rotation of rotor 30 is indicated by direction arrows 50. At each motor state rotor 30 contacts only two of the three stator contact surfaces 47A, 47B, 47C. A given motor state for skipping motor 20 may be designated by a label that includes reference to the two stator contact surfaces that rotor 30 contacts in the given motor state. Additionally, the label indicates a direction of a magnetic B-field generated by electromagnets that are magnetized during a skip that brought the motor to the given motor state and into contact with the two stator contact surfaces of the given motor state. As discussed below, a superscript, +, is used to indicate a clockwise direction of the magnetic B-field and a superscript, -, is used to indicate a counterclockwise direction of the magnetic B-field.

For example, FIG. 2A schematically shows skipping motor 20 in a motor state in which rotor 30 contacts stator contact surfaces 47B and 47A of electromagnets 41B and 41A and is distanced from stator contact surface 47C. A magnetic B-field generated by magnetizing electromagnets 41B and 41A to bring rotor 30 into contact with stator contact surfaces 47B and 47A in accordance with an embodiment is indicated by a dashed line 49 and associated arrows indicating direction of the magnetic B-field along the dashed line. Dashed line 49 and its associated direction arrows may be referred to as a field line 49 or B-field line. For the motor state of skipping motor 20 shown in FIG. 2A field line 49 indicates that the magnetic B-field has a counterclockwise direction. A label (B-A)⁻, in which the minus sign indicates a counterclockwise B-field direction may designate the motor state in FIG. 2A. For a motor state in which rotor 30 contacts stator contact surfaces 47B and 47A but the direction of magnetic B-field is clockwise, as shown for example in FIG. 2D, the motor state is designated (B-A)⁺, in which the plus sign is used to indicate that the B-field has a clockwise direction.

It is noted that whereas the motor state shown in FIG. 2A is labeled (B-A)⁻, it could just as well have been labeled (A-B)⁻, and that the motor state shown in FIG. 2D could have been labeled (A-B)⁺ instead of (B-A)⁺. However, as illustrated for example in FIGS. 2A-2F, any two first and second sequential motor states of skipping motor 20 share a same stator contact surface, and that in skipping from the first motor state to the second motor state, rotor 30 remains in contact with and pivots about the shared stator contact surface between the motor state before the skip and the

motor state after the skip. For clockwise rotation of rotor 30 between a first motor state to a second motor state, for example as shown in the sequence of figures from FIG. 2A to FIG. 2F, the pivot contact surface of stator 40 is the most counterclockwise stator contact surface of the two stator contact surfaces that rotor 30 contacts in the first motor state. For counterclockwise rotation of rotor 30, the pivot surface is the most clockwise pivot surface of the two stator contact surfaces that rotor 30 contacts in the first motor state. For convenience of presentation a label of a given motor state that references the two stator contact surfaces that rotor 30 contacts in the given motor state may recite first the contact surface about which the rotor pivots to reach the given motor state.

For example, for clockwise rotation of rotor 30 illustrated in FIGS. 2A-2F, rotor 30 pivots about contact surface 47A to rotate clockwise from the motor state (B-A)⁻ shown in FIG. 2A to contact stator contact surface 47C to reach the motor state shown in FIG. 2B in which rotor 30 contacts stator contact surfaces 47A and 47C. A label designating the motor state shown in FIG. 2B references the contact surfaces, by reciting first A rather than C, and labels the state (A-C)⁺ rather than (C-A)⁺. Similarly, since the label (B-A)⁻ for the motor state shown in FIG. 2A, recites B before A, the label assumes that the state in FIG. 2A resulted from a clockwise pivot of rotor 30 about stator surface 47B, rather than a counterclockwise pivot about stator surface 47A.

It is noted that in rotating between the motor state (B-A)⁻ to the motor state (A-C)⁺ of skipping motor 20, direction of magnetic B-field reverses from counterclockwise to clockwise. In general, in skipping from a first motor state to a second motor state, in accordance with an embodiment, direction of magnetic B-field in the second motor state is reversed relative to a direction of the magnetic B-field in the first motor state. As a result, for a skipping motor having an odd number of electromagnets in accordance with an embodiment, direction of magnetization of electromagnets effecting contact of the motor's rotor with a given pair of stator contact surfaces reverses every time the rotor rotates in a same clockwise or counterclockwise direction through all the motor states of the motor to return and contact the given pair of stator contact surfaces. For a skipping motor having an even number of electromagnets, the direction of magnetic B-field, which may be clockwise or counterclockwise depending upon initial start-up conditions of the motor, remains the same each time the rotor cycles clockwise or counterclockwise through all the motor states to return to a same pair of stator contact surfaces.

Details and aspects of the clockwise skipping motion of rotor 30 and magnetization of stator electromagnets 41A, 41B, and 41C that generates the clockwise skipping motion of skipping motion shown in FIGS. 2A-2F are discussed below. It should be noted that whereas FIGS. 2A-2F illustrate clockwise rotation and skipping of skipping motor 20 from any motor state to a next motor state, skipping motor 20 may of course rotate counterclockwise and skip counterclockwise from any motor state to a next motor state, as may be realized when considering FIGS. 2A-2F in reverse order from FIG. 2F to FIG. 2A. FIG. 2G discussed below shows a state diagram that schematically illustrates both clockwise and counterclockwise rotation of skipping motor 20, in accordance with an embodiment of the disclosure.

To bring skipping motor 20 to motor state (B-A)⁻ shown in FIG. 2A controller 22 drives currents in coils 45A and 45B to magnetize electromagnets 41A and 41B and induce magnetization in rotor 30 that attracts the rotor to contact stator contact surfaces 47A and 47B of electromagnets 41A

and 41B. The controller drives current in coil 45C to create a coercive "H_c-field" force and demagnetize electromagnet 41C and substantially nullify any remanent, "B_r-field" magnetic flux in core 43C of the electromagnet so that rotor 30 is released from and not attracted to contact surface 47C of electromagnet 41C. The direction of current in electromagnet 41C that creates the H_c-field is reversed to the last direction of H-field and B-field in electromagnet 41C and depends on which was the previous state of the skipping motor 20 before the state shown in FIG. 2A.

In accordance with an embodiment controller 22 controls direction of current through activation coils of a pair of electromagnets to which rotor 30 is attracted in a given motor state so that if a direction of magnetization in one of the pair of electromagnets points in a direction "inward" toward center C_S (FIGS. 1-2F) of stator 40, direction of the magnetization in the other electromagnet of the pair points "outward" away from the center. Controller 22 controls direction of magnetization of a given electromagnet 41A, 41B, and 41C by controlling clockwise or counterclockwise sense of current that the controller generates in activation coil 45A, 45B, and 45C respectively of the electromagnet, relative to a direction extending through the activation coil and facing center C_S of stator 40. If, facing center C_S, current in the activation coil is in the clockwise sense, direction of magnetization and magnetic B-field of the given electromagnet is inward, toward the center C_S. If on the other hand sense of the current in the activation coil is counterclockwise, direction of magnetization and magnetic B-field in the given electromagnet is outward from the center.

As a result of magnetizing a pair of electromagnets in opposite directions in accordance with an embodiment, magnetization of electromagnets 41A, 41B, and 41C for a given motor state provides a relatively low reluctance path indicated by the contour of field line 49 through rotor 30 and a portion of support ring 42 of stator 40 for flux of a magnetic B-field that the magnetization generates.

To cause rotor 30 to skip and rotate clockwise from motor state (B-A)⁻ shown in FIG. 2A to a next, clockwise, motor state (A-C)⁺ shown in FIG. 2B, controller 22 substantially simultaneously: 1) drives a current to demagnetize electromagnet 41B to nullify force between the rotor and stator contact surface 47B; 2) drives activation coil 45C with a counterclockwise current to magnetize electromagnet 41C with an H-field, also referred to as a negative H_c field, in a direction pointing away from center C_S and attract rotor 30 to stator contact surface 47C; and 3) drives activation coil 45A with a clockwise current to magnetize electromagnet 41A with an H-field, also referred to as a positive H_c field, pointing in a direction towards center C_S. As a result, rotor 30 remains in contact with and pivots about stator contact surface 47A, to rotate and contact stator contact surface 47C and rotate skip to motor state (A-C)⁺. Controller 22 operates to demagnetize electromagnet 41B by pulsing activation coil 45 with a clockwise demagnetizing current pulse that generates a magnetic H-field, conventionally often referred to as a coercive magnetic H-field denoted H_c, to substantially null remanent magnetization B_r field of core 43B resulting from hysteresis of the material from which the core, rotor and ring are formed. A block arrow 51 in core 43B in FIG. 2B indicates that in skipping clockwise from motor state (B-A)⁻ to motor state (A-C)⁺, the direction of the coercive field H_c that the demagnetization current pulse generates in core 43B needs to be towards center C_S to remove remanent B-field remaining from B-field 49 shown for the previous motor state shown in FIG. 2A. Magnetic B-field flux for motor state (A-C)⁺ represented by field line 49 in FIG. 2B is

directed clockwise. It is noted that for a counterclockwise rotation of rotor **30**, that is from FIG. 2C to 2B, the demagnetization field H_c -field, arrow **51** in FIG. 2B, would be aimed outward, away from center C_S , and not inward toward C_S as shown in FIG. 2B for clockwise rotation of rotor **30**.

FIG. 2C schematically shows skipping motor **20** in motor state (C-B)⁻ after rotor **30** pivots about stator contact surface **47C** from motor state (A-C)⁺ to rotate and contact stator contact surface **47B** similarly to the way in which the rotor skipped and rotated from motor state (B-A)⁻ to motor state (A-C)⁺. FIG. 2D schematically shows skipping motor **20** having skipped and rotated from motor state (C-B)⁻ shown in FIG. 2C to a motor state (B-A)⁺, in which rotor **30** contacts the same contact surfaces **47A** and **47B** that the rotor contacts in motor state (B-A)⁻ schematically shown in FIG. 2A. However, in motor state (B-A)⁺ magnetic B-field flux represented by field line **49** is clockwise, opposite to the counterclockwise direction of B-field flux for motor state (B-A)⁻. FIGS. 2E and 2F schematically show skipping motor **20** after rotor **30** has rotated respectively to motor states (A-C)⁻ and (C-B)⁺.

FIG. 2G shows a directed state graph **1000** that schematically illustrates both clockwise and counterclockwise rotation of skipping motor **20** shown in FIGS. 1-2F, in accordance with an embodiment of the disclosure. Each motor state of skipping motor **20** is indicated by an ellipse **1002** labeled with the motor state and a figure of FIGS. 2A-2F in which the motor state is shown. Edge arrows **1004** directed clockwise indicate clockwise transitions or skips between states for clockwise rotation of rotor **30**. Each clockwise edge is labeled with directions of the magnetic H-fields that are generated by currents in the electromagnets **241A**, **241B**, and **241C** to produce the skip. IN and OUT refer to H-fields pointing respectively towards and away from center C_S of stator **40** of skipping motor **20**. Similarly edge arrows **1006** directed counterclockwise and labeled with magnetic H-field directions indicate counterclockwise skips between states for counterclockwise rotation of rotor **30**.

FIG. 3 schematically shows ON-OFF activation sequence timelines T_A , T_B , and T_C in accordance with which controller **22** respectively magnetizes and demagnetizes electromagnets **41A**, **41B**, and **41C** as a function of time to generate the clockwise, skipping rotation of rotor **30** of skipping motor **20** shown in FIGS. 2A-2F, in accordance with an embodiment of the disclosure. Activation sequence timelines T_A , T_B , and T_C schematically show electric currents I_A , I_B , and I_C that controller **22** drives through activation coils **45A**, **45B**, and **45C** of electromagnets **41A**, **41B**, and **41C** respectively as a function of time to magnetize and demagnetize the electromagnets assuming continuous clockwise motion of rotor **30**. Time along each timeline T_A , T_B , and T_C is indicated along an abscissa labeled "t". Magnitude of a current driven through a given electromagnet is indicated by displacement from the abscissa along an ordinate labeled "I" of the timeline.

In an embodiment, for continuous clockwise rotation of rotor **30** each current I_A , I_B , and I_C may comprise a train of current pulses **101** of alternating polarity and optionally same duration 2τ . A time period τ is a time it takes for skipping motor **20** to skip from a first motor state to a next second motor state plus a time, a dwell time, it remains in the second motor state after skipping and until it begins a skip to a next, third motor state. A skipping plus dwell time for a particular motor state may be referred to as a cycle period, or period of the motor state. A positive current pulse **101** lies above its associated time abscissa t and represents a clock-

wise current pulse driven through a coil **45A**, **45B**, and **45C** of an electromagnet **41A**, **41B**, and **41C** respectively that magnetizes the electromagnet in a direction pointing inwards toward center C_S (FIGS. 1-2F) of stator **40**. A negative current pulse **101**, lies below its associated time abscissa t , and represents a counterclockwise current pulse that controller **22** drives through a coil **45A**, **45B**, and **45C** of an electromagnet **41A**, **41B**, and **41C** respectively to magnetize the electromagnet in a direction pointing outwards, away from center C_S of stator **40**. Each current pulse **101** is followed by a small demagnetizing current pulse **103**, that creates an H_c -field, schematically represented by block arrows **51** in FIGS. 2A-2F, having direction opposite to that of the H and B-fields created by the given current pulse **101** that the demagnetizing pulse follows. The coercive H_c -field substantially nullifies remanent B_R -field remaining after the given current pulse **101** has ended in the core of the electromagnet to which the given current pulse **101** was applied and in portions of the rotor and support ring. Demagnetizing current pulse **103** may be lower in magnitude and/or shorter in duration than current pulse **101** that it follows because a remanent magnetic B_R -field remaining after cessation of the current pulse is lower than the B-field generated by the current pulse.

Whereas electromagnet currents I_A , I_B , and I_C in an embodiment have optionally substantially same form, to provide rotation of rotor **30** controller **22** controls their respective time developments as functions of time with different phases, or time displacements relative to a common time origin. For example, in an embodiment of the disclosure to provide clockwise rotation of rotor **30** and the motor states as shown in FIGS. 2A-2F relative to an arbitrary time origin to in FIG. 3A, a first positive current pulse in electromagnet current I_A is delayed by a time $4t$, electromagnet current I_B is not delayed, and electromagnet current I_C is delayed by a time $2t$. For the configuration of electromagnet currents shown in FIG. 3A skipping motor **20** cycles continuously through motor states (B-A)⁻, (A-C)⁺, (C-B)⁻, (B-A)⁺, (A-C)⁻, and (C-B)⁺, shown in FIGS. 2A-2F with the skipping motor skipping to and remaining in each motor state for a motor state period equal to about t , that is, half a pulse width of a current pulse **101**. The motor states and their respective cycle periods are shown at the bottom of FIG. 3A below segments of electromagnet currents I_A , I_B , and I_C along activation sequence timelines T_A , T_B , and T_C that generate the motor states. The cycle periods of the motor states and their respective corresponding segments of electromagnet currents I_A , I_B , and I_C , are demarked by vertical dashed lines labeled by the onset times of the motor states. It is noted that in FIG. 3A the de-magnetization current pulse **103** is shorter than t , as in practice, this current pulse may be need only until the remanent magnetic B-field is zeroed out. However, for convenience of control, current pulse **103** may continue for the full time T of a state.

It is noted that for the continuous rotation assumed for ON-OFF activation sequence timelines T_A , T_B , and T_C shown in FIG. 3A electromagnet currents that attract rotor **30** to stator contact surfaces of stator **40** for a given motor state are continuously on from the beginning of a skip to the given motor state to a time at which skipping motor **20** is controlled to skip to a next state. That is, the "attractive" electromagnet currents are continuously on for the duration of the given motor state cycle period. In an energy saving mode of operation of skipping motor **20**, electromagnet currents that attract rotor **30** to stator contact surfaces of stator **40** for a given motor state are stopped before the end of the motor state cycle period. The rotor is held in the given

motor state by a remanent magnetic B-field from the time the currents are stopped until the end of the motor state cycle period. FIG. 3B shows a mode of operation of skipping motor 20 for which "attractive" electromagnet current pulses are shorter than motor state cycle periods. Current pulses 101 shown in FIG. 3A are replaced by current pulses 101a and 101b in FIG. 3B.

It is further noted that when motor 20 is stopped it may be held in a last motor state by remanent B-fields or by maintaining the electromagnet currents that attracted rotor 30 to stator contact surfaces of stator 40 for the last motor state. For hardened soft magnetic steel AISI 1020 it was found that remanent flux held the rotor in place for 3 months, with torque similar to a torque with energized activation coils. With hardened hard magnetic steel such as AISI 1090 with 0.9% carbon, the remanent flux and the holding torque is expected to be higher. If it is desired that the motor be free to rotate when stopped, a pulse of current may be used in the last two activation coils to be energized, to cancel remanent flux. Alternatively, clockwise and counterclockwise electromagnet current pulses with diminishing amplitude may be used to demagnetize the electromagnets. Alternatively, cores of the electromagnets may be formed from a silicon electrical steel to minimize remanent flux.

FIGS. 3C-3E show another, alternative mode of operation, optionally referred to as Smooth-Skip, by which controller 22 may operate and control skipping motor 20 to provide efficient and smooth skipping between motor states of the skipping motor, in accordance with an embodiment of the disclosure.

For convenience of presentation Smooth-Skip is discussed with reference to skipping motor 20 skipping from motor state (B-A)⁻ to motor state (A-C)⁺ in accordance with an embodiment of the disclosure. Stator 40 and rotor 30 of the skipping motor are assumed to be formed from a material having a same permeability, $\mu_o \mu(m)$, where μ_o is the permeability of free space and $\mu(m)$, is the relative permeability of the material. Relevant B-fields in the figures are schematically represented by respective field lines with arrows on the field lines indicating directions of the associated B-fields along the field lines. Each B-field and associated field line are referenced by an alphanumeric label indicating cores, 43A, 43B, and/or 43C of motor 20 through which the B-field passes, a direction of the B-field in rotor 30, and which activation coil in skipping motor 20 is excited to generate the B-field.

For example, consider motor state (B-A)⁻ of motor 20 schematically shown in FIG. 3C, in which rotor 30 is held in contact with contact surfaces 47A and 47B and assume that a positive current I_o excites activation coil 45A of core 43A. The excitation generates a magnetic B-field 49A*B along a field line also referenced by 49A*B in directions indicated by arrows on the field line. The asterisk over the letter A in label 49A*B indicates that B-field 49A*B is generated by current through activation coil 45A, and the order of the letters, A before B, in the label indicates that, in rotor 30, the B-field points from core A towards core B. Current I_o also generates a B-field 49A*C δ in directions indicated by arrows along a field line 49A*C δ that passes through cores 43A and 43C. The δ subscripting C indicates that in motor state (B-A)⁻ there is an air gap of width δ along field line 49A*C δ between rotor 30 and core 43C. B-fields 49A*B and 49A*C δ combine in core 43A to form a B-field 49A*BC δ =49A*B+49A*C δ . B-field 49A*C δ is shown as a dashed line to indicate that the magnetic B-field 49A*C δ is much weaker than B-field 49A*B due to the air gap S along field line 49A*C δ .

An attractive force in motor state (B-A)⁻ that B-field 49A*BC δ generates between rotor 30 and core 43A of stator 40 is proportional to magnetic flux generated by B-field 49A*BC δ that passes through contact surface 47A of core 43A. Assume that activation coil 45A as well as activation coils 45B and 45C comprise N activation coil windings around their respective cores 43A, 43B, and 43C and contact surfaces 47A, 47B, and 47C have a same area A. Let $\Phi(A*BC_\delta)$ represent magnetic flux generated by B-field 49A*BC δ through area A and let $\Phi(A*B)$ and $\Phi(A*C_\delta)$ represent magnetic flux of B-fields 49A*B and 49A*C δ through A respectively. Then

$$\Phi(A*BC_\delta) = \Phi(A*B) + \Phi(A*C_\delta) = N \cdot I_o / \mathcal{R}(A*BC_\delta), \quad (1)$$

where $\mathcal{R}(A*BC_\delta)$ is a reluctance of materials along respective magnetic circuits through which B-fields 49A*B and 49A*C δ that contribute to $\Phi(A*BC_\delta)$ pass. The magnetic circuits lie substantially along field lines 49A*B and 49A*C δ respectively and may be referred to as magnetic circuits 49A*B and 49A*C δ . The magnetic circuits share substantially similar same length magnetic path lengths, which may be considered a common path length, through core 43A and rotor 30 along which B-fields 49A*B and 49A*C δ "leave" core 43A. The common path length is in series with separate, parallel magnetic circuit path lengths that respectively pass through cores 43B and 43C and different portions of stator 40 along which the B-fields return to core 43A.

To simplify estimation of magnetic flux $\Phi(A*BC_\delta)$ it is assumed that $\Phi(A*B)$ and $\Phi(A*C_\delta)$ may be approximated independently so that $\Phi(A*B)$ is considered generated only by an H-field produced by current I_o and a reluctance $\mathcal{R}(A*B)$ of magnetic circuit 49A*B, and $\Phi(A*C_\delta)$ is considered generated only by the same H-field and a reluctance $\mathcal{R}(A*C_\delta)$ of magnetic circuit 49A*C δ . The assumption is tenable because generally the reluctance of the common magnetic circuit path length is much lower than that of the separate magnetic path lengths and because of the air gap S along magnetic circuit 49A*C δ , reluctance $\mathcal{R}(A*C_\delta)$ of magnetic circuit 49A*C δ is much higher than reluctance $\mathcal{R}(A*B)$ of magnetic circuit 49A*B. Under these assumptions magnetic fluxes $\Phi(A*B)$ and $\Phi(A*C_\delta)$ may be written

$$\Phi(A*B) \cong N \cdot I_o / \mathcal{R}(A*B), \text{ and} \quad (2)$$

$$\Phi(A*C_\delta) \cong N \cdot I_o / \mathcal{R}(A*C_\delta) \quad (3)$$

Let $L(A*B)$ represent a length of the contour of magnetic circuit 49A*B, as schematically illustrated by field line 49A*B, and $L(A*C_\delta)$ represent a length of the contour of magnetic circuit 49A*C as schematically illustrated by field line 49A*C δ . Magnetic circuit 49A*B may be referred to as a "continuous circuit" in a sense that the contour of magnetic circuit of length $L(A*B)$ passes only through the material, for example a metal such as steel, from which stator 40 and rotor 30 are formed. On the other hand, magnetic circuit 49A*C δ may be referred to as an "interrupted circuit", in a sense that a portion of the path length $L(A*C_\delta)$ of the contour of magnetic circuit 49A*C δ , includes and is "interrupted by" air gap 6. Contour lengths $L(A*B)$ and $L(A*C_\delta)$ have substantially a same physical length L_o and if not for

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air gap S, the circuit contours of magnetic circuit **49A*B** and magnetic circuit **49A*C_δ** would, except for position, be substantially identical. In view of the above, the reluctances $\mathcal{R}(A*B)$ and $\mathcal{R}(A*C_\delta)$, may be estimated by the expressions:

$$\mathcal{R}(A*B) \cong L_O/(\mu_O \cdot \mu(m)_r A) \text{ and} \quad (4)$$

$$\mathcal{R}(A*C_\delta) \cong (L_O - \delta)/(\mu_O \cdot \mu(m)_r A) + \delta/(\mu_O \cdot \mu(\delta)_r A) \quad (5)$$

In expressions (4) and (5) a cross-section area of magnetic circuits **49A*B** and **49A*C_δ** along their respective path lengths is assumed to be constant and equal to the area A of contact surfaces **47A** and **48A**. In expression (5), $\mu(\delta)_r$ is the relative permittivity of air. Since $\mu(\delta)_r$ is about equal to 1 and $L_O \gg \delta$, expression (5) may be written as:

$$\mathcal{R}(A*C_\delta) \cong L_O/(\mu_O \cdot \mu(m)_r A) + \delta/(\mu_O \cdot A) \quad (6)$$

Using expressions (4) and (6) for reluctances $\mathcal{R}(A*B)$ and $\mathcal{R}(A*C_\delta)$ in expressions (2) and (3), fluxes $\Phi(A*B)$ and $\Phi(A*C_\delta)$, may be written:

$$\Phi(A*B) \cong N \cdot I_O(\mu_O \cdot A)/(L_O/\mu(m)_r) = K/(L_O/\mu(m)_r), \text{ and} \quad (7)$$

$$\Phi(A*C_\delta) \cong N \cdot I_O(\mu_O \cdot A)/(L_O/\mu(m)_r + \delta) = K/(L_O/\mu(m)_r + \delta), \quad (8)$$

where $K=N \cdot I_O(\mu_O \cdot A)$. $\Phi(A*BC_\delta)$ may therefore be given by an expression,

$$\Phi(A*BC_\delta) = \quad (9)$$

$$\Phi(A*B) + \Phi(A*C_\delta) \cong K/(L_O/\mu(m)_r) + K/(L_O/\mu(m)_r + \delta) = K[1/(L_O/\mu(m)_r) + 1/(L_O/\mu(m)_r + \delta)]$$

Similarly, from symmetry, assuming that in motor state (B-A)⁻ a current equal to, $-I_O$, flows through activation coil **45B**, the current generates a magnetic flux $\Phi(AC_\delta B^*)$ which is a sum of fluxes $\Phi(AB^*) + \Phi(C_\delta B^*)$. Flux $\Phi(AB^*)$ passes through cores **43A** and **43B** along magnetic circuit **49AB*** in a same direction as flux $\Phi(A*B)$ passes through activation coils **43A** and **43B** along magnetic circuit **49A*B**. Flux $\Phi(C_\delta B^*)$ passes through cores **43C** and **43B** along magnetic circuit **49C_δB*** in a direction through core **43C** opposite to a direction in which flux $\Phi(A*C_\delta)$ passes through core **43C**. In core **43C** flux $\Phi(A*C_\delta)$ and flux $\Phi(C_\delta B^*)$ substantially cancel each other. Flux $\Phi(AC_\delta B^*)$ may be estimated by an expression similar to expression (9):

$$\Phi(AC_\delta B^*) = \Phi(AB^*) + \Phi(C_\delta B^*) = K[1/(L_O/\mu(m)_r) + 1/(L_O/\mu(m)_r + \delta)] \quad (10)$$

$\Phi(AC_\delta B^*)$ and $\Phi(A*BC_\delta)$ are magnetic fluxes generated by currents $-I_O$ and I_O in activation coils **45B** and **45A** respectively for a same magnetic circuit that lies along both field lines **49A*B** and **49AB***. The fluxes pass through each of cores **43A** and **43B** in a same direction and add to provide a compound flux $\Phi(A*C_\delta B^*)$. The compound flux may be approximated by an expression:

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$$\Phi(A*C_\delta B^*) = \quad (11)$$

$$\Phi(A*BC_\delta) + \Phi(AC_\delta B^*) = 2K[1/(L_O/\mu(m)_r) + 1/(L_O/\mu(m)_r + \delta)].$$

The compound flux operates when skipping motor **20** is in motor state (B-A)⁻ to provide magnetic attraction of rotor **30** to both cores **43A** and **43B** that hold the rotor securely to contact surfaces **47A** and **47B**.

FIG. **3D** schematically illustrates initiating a skip from motor state (B-A)⁻ shown in FIG. **3C** to motor state (A-C)⁺, schematically shown in FIG. **3E**, in accordance with an embodiment of the disclosure. Initiating the skip involves controlling current in activation coil **45B** to reverse direction and increase from $-I_O$ to a positive coercive current I_c and generating a current $-I_O$ in activation coil **45C**. Coercive current I_c is determined to produce a coercive H_c -field represented by a block arrow **51** in core **43B** that operates to nullify remanent B_R -field remaining from current $-I_O$ that flowed through activation coil **45B** in motor state (B-A)⁻ and produced magnetic flux $\Phi(AB^*)$ which attracted rotor **30** to contact surface **47B**. Current $-I_O$ generated in activation coil **45C** operates to produce fluxes $\Phi(AC^*\delta)$ and $\Phi(BC^*\delta)$ along interrupted magnetic circuits **49AC_δ*** and **49BC_δ*** respectively and a flux $\Phi(ABC^*\delta)$ through core **43C** that operates to attract rotor **30** to **43C**, move the rotor away from core **43B**, and rotate the rotor clockwise.

From an analysis similar to that described above for estimating magnetic fluxes, for the state shown in FIG. **3D**, fluxes $\Phi(AC^*\delta)$, $\Phi(BC^*\delta)$, and $\Phi(ABC^*\delta)$ may be approximated by expressions:

$$\Phi(AC_\delta^*) = \Phi(BC_\delta^*) \cong K/(L_O/\mu(m)_r + \delta), \text{ and} \quad (12)$$

$$\Phi(ABC_\delta^*) = \Phi(AC_\delta^*) + \Phi(BC_\delta^*) = 2K/(L_O/\mu(m)_r + \delta). \quad (13)$$

Whereas increasing $-I_O$ in activation coil **47B** to positive coercive current I_c operates to cancel magnetic flux $\Phi(AB^*)$ that attracts rotor **30** to core **43B** and free the rotor from core **43B** so that the rotor can rotate to motor state (A-C)⁺, current I_O in activation coil **45A** remains to continue generating magnetic flux $\Phi(A*B)=K/(L_O/\mu(m)_r)$ and hold rotor **30** to core **43A** so that the rotor may rotate to motor state (A-C)⁺. However, $\Phi(A*B)$ also operates to attract rotor **30** to core **43B** with a relatively strong force and “hamper” rotation of the rotor to motor state (A-C)⁺.

In accordance with an embodiment to reduce the attractive force generated by magnetic flux $\Phi(A*B)$ that hampers rotation of rotor **30**, when initiating skipping to motor state (A-C)⁺, current in activation coil **45A** may be reduced so that flux $\Phi(A*B)$ has a magnitude substantially equal to that of flux $\Phi(BC^*\delta)$ and direction in core **43B** opposite to that of $\Phi(BC^*\delta)$, as a result of which fluxes $\Phi(A*B)$ and $\Phi(BC^*\delta)$ cancel each other in core **43B**. The constraint of equality may be satisfied if current in activation coil **45A** is reduced to a skip current I_S so that

$$\Phi(A*B) \cong N \cdot I_S(\mu_O \cdot A)\mu(m)_r/L_O = \quad (14)$$

$$-\Phi(BC_\delta^*) \cong -N \cdot I_O(\mu_O \cdot A)/(L_O/\mu(m)_r + \delta),$$

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which leads to a requirement that

$$I_S/I_O = -(L_O/\mu(m)_r)/(L_O/\mu(m)_r + \delta) = -1/(1 + \delta \cdot \mu(m)_r/L_O). \quad (15)$$

It is noted that in expression (15) the minus sign accounts for the direction of current in activation coil **45C** being opposite to that of the direction of current in activation coil **45A**. A positive current is a current that generates a magnetic B-field that points into rotor **30** and a negative current is a current that generates a magnetic B-field that points out from the rotor.

By substantially satisfying expression (15) rotor **30** may be substantially free of force holding the rotor to core **43B** and the rotor free to easily disengage from core **43B** and begin rotating to state (A-C)⁺.

In an embodiment, construction of a skipping motor is configured to provide for reducing attractive force between core **43B** and rotor **30** to initiate a relatively fast unhampered skip from motor state (B-A)⁻ to motor state (A-C)⁺ without reducing current in coil **45A**. The configuration provides that upon initiating the skip under conditions for which excitation current in coils **45A** and **45C** are substantially the same, a force of magnetic attraction between rotor **30** and core **43B** is less than a force of magnetic attraction between the rotor and core **43C**. In symbols the condition may be written,

$$|\Phi(A * B) + \Phi * BC_\delta^*| < |\Phi(A * C_\delta) + \Phi(AC_\delta^*) + \Phi(BC_\delta^*)|. \quad (16)$$

Since it is assumed that magnitudes of currents in coils **45A** and **45C** are substantially equal, condition (16) may be written

$$|1/(L_O/\mu(m)_r) - 1/(L_O/\mu(m)_r + \delta)| < \quad (17)$$

$$|1/(L_O/\mu(m)_r + \delta) + 2/(L_O/\mu(m)_r + \delta)|, \quad (18)$$

which leads to a requirement that

$$\delta < 3L_O/\mu(m)_r. \quad (18)$$

By way of a numerical example, if $L_O=100$ mm, and $\mu(m)_r=1000$, condition (18) becomes

$$\delta < 0.3 \text{ mm} \quad (19)$$

In an embodiment, after rotor **30** disengages from core **43B** and rotates about halfway to state (A-C)⁺ current in core **43A** may be returned to current I_O to accelerate rotation of the rotor to state (A-C)⁺ and hold the rotor to cores **43A** and **43C** upon reaching state (A-C)⁺ shown (A-C)⁺ shown in FIG. **3E**. In state (A-C)⁺ magnetic fluxes $\Phi(A * C)$ and $\Phi(AC^*)$ along continuous magnetic circuits **49A * C** and **49AC *** combine to securely hold rotor **30** to cores **43A** and **43C** and magnetic fluxes **49A * B** and **49BC *** cancel in core **43B**.

FIGS. **4A** and **4B** schematically illustrate parameters of skipping motor **20** shown in FIGS. **1-2F** that may be used to determine a magnitude of an angle, a skip angle by which the

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motor rotates in skipping from a first motor state to a next, second motor state of the motor in accordance with an embodiment of the disclosure. FIGS. **4A** and **4B** show highly schematic drawings of skipping motor **20** in which features of skipping motor **20** shown in FIGS. **1-2F** are represented by idealized geometric entities labeled by the labels that respectively label the features in FIGS. **1-2F**. Stator **40** is represented by a bold line circle labeled **40** and stator contact surfaces **47A**, **47B** **47C**, are represented by solid circles labeled **47A**, **47B** **47C**. Stator **40** has a center C_S and radius R_S . Rotor **30** is represented by a thin line circle **30**, having a center C_R and radius R_R . FIG. **4A** schematically shows skipping motor **20** in motor state (C-B)⁻ shown in FIG. **2C**. FIG. **4B** shows skipping motor **20** in motor state (A-C)⁺ shown in FIG. **2B** that preceded motor state (C-B)⁻ as rotor **30** rotated clockwise. FIG. **4B** also shows skipping motor **30** in motor state (C-B)⁻. In FIG. **4B** for motor state (A-C)⁺ rotor **30** is represented by a solid line circle. For motor state (C-B)⁻ in FIG. **4B** rotor **30** is shown as a dashed line circle. The skip angle between motor state (A-C)⁺ and motor state (C-B)⁻ is shown in FIG. **4B** as an angle Φ .

In motor state (C-B)⁻ shown in FIG. **4A** stator **40** and rotor **30** share a same chord labeled for convenience by its length "L". Chord L subtends an angle Θ_S at center C_S of stator **40**. In general, Θ_S is equal to $2\pi/N$ where N is a number of stator contact surfaces that the stator comprises. For skipping motor **20**, $N=3$ and $\Theta_S=2\pi/3=120^\circ$. Chord L subtends an angle Θ_R at center C_R of rotor **30**. Distance of center C_S from chord L is represented by a height H_S and distance of center C_R from chord L is represented by a height H_R . Let $\Delta H=(H_S-H_R)$. Then

$$\Delta H = (H_S - H_R) = [R_S \cos(\Theta_S/2) - R_R \cos(\Theta_R/2)]. \quad (16)$$

Skipping angle Φ between motor state (A-C)⁺ and motor state (C-B)⁻ shown in FIG. **4B** may be written,

$$\Phi = 2 \arctan \left[\left(\Delta H / R_R \right) \sin(\pi/3) \right]. \quad (17)$$

For $(R_S - R_R) \ll R_R$ and as a result $\Delta H \ll R_R$ expression (2) becomes,

$$\Phi \cong 2(\Delta H / R_R) \sin(\pi/3) = 2 \left[\left((R_S / R_R) \cos(\Theta_S/2) - \cos(\Theta_R/2) \right) \sin(\pi/3) \right]. \quad (18)$$

FIGS. **5A-5C** schematically show configuration of skipping motors in accordance with embodiments of the disclosure that are variations of skipping motor **20**.

FIG. **5A** schematically shows a skipping motor **200** comprising a rotor **230** having a center C_R and radius R_R seated inside a stator **240** in accordance with an embodiment of the disclosure. Stator **240** comprises, optionally three, electromagnets **241A**, **241B**, and **241C**, having cores **243A**, **243B**, **243C** that are supported on a base plate **244** and are surrounded respectively by conductive activation coils **245A**, **245B**, and **245C**. Cores **243A**, **243B**, and **243C** have respective extensions **248A**, **248B**, and **248C**, and each core has a shelf **246** and a spacer **249** seated on the shelf on which rotor **240** rests. In the perspective of FIG. **5A** only shelf **246** and spacer **249** of electromagnet **241C** are shown. Extensions **248A**, **248B**, and **248C** of electromagnets **241A**, **241B**, and **241C** may have concave, crescent contact surfaces

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247A, 247B, and 247C, optionally having a radius of curvature R_S greater than radius R_R of rotor 230. Crescent contact surfaces 247A, 247B, and 247C optionally exhibit threefold rotational symmetry and lie on a same circular cylindrical surface (not shown) having radius R_S . Spacers 249 are advantageously made from non-ferromagnetic material, and operate to distance rotor 230 from shelves 246, in order to reduce the magnetic force between rotor 230 and shelves 246. In an embodiment, spacers 249 are made of low friction material, such as teflon, to lower friction forces between rotor 230 and shelves 246.

FIG. 5B schematically shows a skipping motor 300 optionally comprising an annular rotor 330 and a stator 340 having, optionally three, electromagnets 341A, 341B, and 341C each having a core 343 supported on a base plate 344. Cores 343 of electromagnets 341A, 341B, and 341C are surrounded respectively by conductive coils 345A, 345B and 345C, and each core is formed having a shelf 346 and a spacer 349 seated on the shelf on which annular rotor 330 rests. In the perspective of FIG. 5A only shelf 346 and spacer 349 of electromagnet 341B are shown. In an embodiment cores 343 of electromagnets 341A, 341B, are formed having convex contact surfaces 347A, 347B, and 347C respectively that are surrounded by rotor 330. Rotor 330 has an internal radius R_R and convex contact surfaces 347A, 347B, and 347C optionally have a radius of curvature R_S and are tangent to a circular cylindrical surface having radius R_S smaller than R_R . Shelves 346 are covered by spacers 349 that reduce friction between shelves 346 and rotor 330. Spacers 349 are advantageously made from non-ferromagnetic material, and operate to distance rotor 330 from shelves 346, in order to reduce the magnetic force between rotor 330 and shelves 346. In an embodiment, spacers 349 are made of low friction material, such as teflon, to lower friction forces between rotor 330 and shelves 346.

FIG. 5C schematically shows a skipping motor 400 comprising a stator 440 having a hub 442 from which electromagnets 441A, 441B, and 441C extend and an annular rotor 430 surrounding the stator in accordance with an embodiment of the disclosure. Electromagnets 441A, 441B, and 441C have cores 443 that are surrounded respectively by conductive coils 445A, 445B, and 445C and have contact surfaces 447A, 447B, and 447C respectively. Skipping motor 400 is substantially an inversion of skipping motor 20 shown in FIGS. 1-2F, and has electromagnets formed on an internal stator surrounded by an annular rotor rather than having electromagnets formed on an external stator surrounding an internal rotor.

By way of a numerical example, a skipping motor constructed in accordance with an embodiment similar to that shown in FIG. 5B, comprised a rotor 330 having inner radius R_R equal to about 36 mm (millimeters), and a stator 340 having electromagnets 341A, 341B, and 341C and radius R_S equal to about 35.65 mm. The skipping motor exhibited a skip angle equal to about $2\pi/400=0.016$ radians or about 1 degree. Another skipping motor having R_R equal to about 36 mm and R_S equal to about 35.89 mm exhibited a skip angle equal to about $2\pi/800=0.008$ radians or about $\frac{1}{2}$ degree.

Cores 343 of electromagnets 341A, 341B, and 341C of the skipping motors in the example were formed from a relatively soft magnetic AISI (American Iron and Steel Institute) 1020 carbon steel with 0.2% carbon. The steel surfaces were blackened with gun blue (selenious acid) and very slightly oiled. Coils 345A, 345B, and 345C, of the electromagnets were APW model FC-4776 electromagnet stock coils, having bobbin length of 16.6 mm, inner diameter of 12.7 mm and outer diameter of 22 mm. The coils had

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1990 turns and were designed for 24 V operation. The motors were controlled by an Ace controller made by Velocio.net and three driver chips TIL293 made by Texas Instruments, with each chip controlling a different coil. A PC based program was used to create run programs for the controller, which allowed to select the skip timing and the number of skips. Due to the controller limitation, the fastest skip time was 10 ms.

A force between a core 343 of an electromagnet 341A, 341B, or 341C and rotor 430 due to driving the electromagnet's coil with 24 volts DC was calculated to be 30 N (Newtons). Assuming a friction coefficient of 0.25 between slightly oiled steel surfaces, a resultant holding torque of rotor 330 provided by contact of the rotor with contact surfaces of cores 343 of two of the electromagnets, was calculated to be 0.27 Nm (Newton-meter). A holding torque of 0.35 Nm was actually measured.

FIGS. 6A and 6B schematically show a cross-section plan view of a skipping motor 500, hereinafter also referred to as a "roloskip" motor, in accordance with an embodiment of the disclosure that exhibits an advantageously smooth, rolling motion in skipping from a first motor state to a second motor state.

Roloskip motor 500 optionally comprises an annular rotor 530 having an inner surface 531 of radius R_R and center C_R surrounding a stator 540, optionally similar to stator 240 shown in FIG. 5B, but having a plurality of N, optionally equal to three, substantially identical, electromagnets 541A, 541B, and 541C. The electromagnets may be configured in a threefold rotationally symmetric geometry about an axis of the stator that intersects and is perpendicular to the cross-section view in FIGS. 6A and 6B at a center C_S of a virtual surface of rotation indicated by a dashed circle 548 having a radius R_S . As described in the discussion below each electromagnet may comprise two contact surfaces and when skipping between a first and a second motor state, rotor 530 "skip-rolls" from one to another of the contact surfaces of a same electromagnet. FIG. 6A shows roloskip motor 500 in a motor state (A-C) in which rotor 530 contacts stator contact surfaces of electromagnets 541A and 541C. FIG. 6B shows roloskip motor 500 after a counterclockwise rotation of rotor 530 to a motor state (C-A), in which rotor 530 contacts stator contact surfaces of electromagnets 541C and 541A.

Electromagnet 541A is formed having a compound surface 542A facing rotor 530 that comprises left and right stator contact surfaces 542A-L, and 542A-R respectively. The left and right contact surfaces are optionally spliced to a same, optionally planar, join surface 542A-J at respective left and right seams. The seams are indicated and referenced by left and right seam points 543A-L and 543A-R respectively at which the seams intersect the plane of FIG. 6A. In an embodiment, left and right contact surfaces 542A-L and 542A-R of electromagnet 541A are mirror image surfaces in a plane indicated by a line labeled 555A that passes through center C_S . Left and right seam points 543A-L and 543A-R lie on circle 548. Join surface 542A-J between seam points 543A-L and 543A-R subtends an angle R at center C_S . Reference to left and right with respect to electromagnet 451A refers respectively to left and right sides of plane 555A when looking outward from center C_S toward the electromagnet. Each left and right contact surface 542A-L and 542A-R of electromagnet 541A has a radius of curvature substantially equal to inner radius R_R of annular rotor 530. Tangents 556A-L and 556A-R to right and left contact surfaces 542A-L and 542A-R at seam points 543A-L and 543A-R respectively, enclose an angle α .

Similarly, electromagnet **541B** has a compound surface **542B** having left and right contact surfaces **542B-L** and **542B-R** having respective radii of curvature equal to R_R . Contact surfaces **542B-L**, and **542B-R** are mirror images of each other in a plane **555B** and are spliced to an optionally planar, join surface **542B-J** at respective left and right seam points **543B-L** and **543B-R**. Tangents (not shown) to left and right contact surfaces **542B-L**, and **542B-R** at seam points **543B-L** and **543B-R** enclose an angle α . And electromagnet **541C** has a compound surface **542C** having left and right contact surfaces **542C-L** and **542C-R** having respective radii of curvature equal to R_R . **542C-L**, and **542C-R** are mirror images of each other in a plane **555C** and are joined at seams **543B-L** and **543B-R** to an optionally planar surface **542C-J**. Tangents (not shown) to left and right contact surfaces **542C-L** and **542C-R** at seam points **543C-L** and **543C-R** enclose an angle α . As in the case of electromagnet **541A**, left and right seam points of electromagnets **541B**, and **541C** lie on circle **548**.

At any given motor state of roloskip motor, a left or right stator contact surface of one of electromagnets **541A**, **541B**, or **541C** and a right or left stator contact surface respectively of another of electromagnets **541A**, **541B**, or **541C** are substantially coincident with portions of inner surface **531** of rotor **530**. For example, as noted above, FIG. **6A** shows roloskip motor **500** in a motor state (A-C) in which rotor **530** contacts stator electromagnets **541A** and **541C** and the figure shows that left and right contact surfaces **542A-L** and **542C-R** are substantially coincident with portions of inner surface **531** of rotor **530**. Similarly, FIG. **6B** shows roloskip motor **500** in motor state (C-B) in which rotor **530** contacts stator electromagnets **541C** and **541B** and the figure shows that left and right contact surfaces **542C-L** and **542B-R** are substantially coincident with portions of inner surface **531** of rotor **530**.

In skipping from a first to a second motor state of roloskip motor **500**, as for example skipping between motor state (A-C) shown in FIG. **6A** and motor state (C-B) shown in FIG. **6B**, rotation of rotor **530** is characterized by an amplitude of nutation D shown in FIG. **6A**. Assuming that rotor **530** does not slip on stator contact surfaces of stator **540** when skipping between motor states, a skip angle between the motor states may be estimated to be equal substantially to the angle α discussed above.

For a roloskip motor similar to roloskip motor **500** but having N electromagnets similar to electromagnets comprised in roloskip motor **500** that are configured in an N -fold symmetric geometry, relationships between a desired skip angle α and geometric parameters of the roloskip motor may be provided by the following expressions:

$$\Theta_S = 2\pi/N; \quad (19)$$

$$D = -(R_S \sin \beta / 2) / \cos \Theta_S; \quad (20)$$

$$\tan \beta = (R_R \sin \alpha - D \sin \Theta_R / 2) / (R_R \cos \alpha - D \cos \Theta_R / 2); \text{ and} \quad (21)$$

$$R_S = [(R_R \sin \alpha - D \sin \Theta_R / 2)^2 + (R_R \cos \alpha - D \cos \Theta_R / 2)^2]^{1/2}. \quad (22)$$

For roloskip motor **500**, $N=3$, and electromagnets **541A**, **541B**, and **541C**, may be pulsed to magnetize and demagnetize the electromagnets to rotate rotor **530** clockwise or counterclockwise similarly to the manner in which electromagnets **41A**, **41B**, and **41C** comprised in skipping motor **20** are pulsed to rotate rotor **30** respectively counterclockwise and clockwise.

FIG. **7** schematically shows another roloskip motor **600** configured to provide a relatively smooth rolling motion between motor states, in accordance with an embodiment of the disclosure. Roloskip motor **600** comprises a rotor **630** and a stator **640** having electromagnets **641A**, **641B** and **641C**, which have respective cores **643A**, **643B** and **643C** formed having stator contact surfaces **647A**, **647B**, and **647C** respectively. Spaces between cores **643A**, **643B** and **643C** are filled with a filler material **641** advantageously formed from a nonferromagnetic wear resistant filler material, such as by way of example, a glass epoxy, polyimide, brass, or aluminum. Advantageously the filler is formed to exhibit relatively low electrical conductivity to moderate eddy current energy loss. To provide smooth and quiet skipping between motor states, stator contact surfaces **647A**, **647B**, and **647C** and a surface **642** of filler **641** lie on a same circular cylindrical surface having a center C_S and radius R_S .

FIGS. **8A** and **8B** schematically show cross-sections of a skipping motor **700** in accordance with an embodiment of the disclosure for which a ratio, a skipping ratio, between a radius of a rotor and a stator that the motor comprises, and thereby a skipping angle of the motor are adjustable.

Skipping motor **700** optionally comprises a stator **740** having four electromagnets **741A**, **741B**, **741C** and **741D** only two of which, electromagnets **741A** and **741C**, that are opposite each other are shown in the figure. Electromagnets **741A**, **741B**, **741C** and **741D** are optionally arranged in a rotationally symmetric configuration having an angle of rotational symmetry equal to $2\pi/4$. The skipping motor comprises a rotor **730** mounted to a set screw stem **750** having a friction reducing bearing **752** on which the rotor rests in accordance with an embodiment of the disclosure. Set screw stem **750** is secured to rotor **730** by an optionally elliptical head **754** of the set screw stem that is held in a socket **732** of the rotor. Dimensions and tolerances of elliptical head **754** and socket **732** are determined to allow nutating skipping rotation of rotor **730**. The set screw stem seats in a threaded hole **743** in a base **744** of stator **740** having threads that match threads of the set screw stem. Electromagnets **741A-741D** are formed having stator contact surfaces **747A-747D** respectively that are portions of a surface of a right circular cone angled in the cross section shown in FIGS. **8A** and **8B** by an angle λ with respect to an axis **745** of the stator. In the cross-section view of FIGS. **8A** and **8B** only stator contact surfaces **747A** and **747C** are shown. An edge surface **733** of rotor **730** is angled with respect to axis **745** optionally by same angle λ with respect to axis **745** as are stator contact surfaces **747A** and **747C**. Rotor **730** has an average radius R_R .

Position of rotor **730** along axis **745** relative to stator contact surfaces **747A** and **747C** may be changed by rotation set screw stem **750** to raise or lower rotor **730** relative to base **744** of stator **740**. Raising rotor **730** relative to base **744** positions the rotor at locations along axis **745** for which a distance between opposite stator contact surfaces **747A** and **747C**, is greater and a radius of the stator is greater. Lowering rotor **730** relative to base **744** positions the rotor at locations along axis **745** for which a distance between opposite stator contact surfaces **747A** and **747C** is smaller and a radius of the stator is smaller. Raising and lowering rotor **730** increases and decreases respectively a radius of stator **740** at the location of the rotor along axis **745** and thereby increases and decreases the skipping ratio, and as indicated by expression 3) respectively increases and decrease the skipping angle of skipping motor **700**. FIG. **8A** schematically shows skipping motor **700** with rotor **730** located at a first height above base **744** for which stator **740**

has an average radius R_{S1} . FIG. 8B shows skipping motor 700 having rotor 730 located at a second height above base 744 for which stator 740 has an average radius $R_{S2} > R_{S1}$. A skipping ratio and skipping angle of skipping motor 700 shown in FIG. 8B are respectively greater than a skipping ratio and skipping angle of skipping motor 700 shown in FIG. 8A.

FIGS. 9A and 9B schematically show cross-sections of a skipping motor 800 in accordance with an embodiment of the disclosure for which an angle between an axis of rotation of a rotor in the skipping motor may be changed relative to an axis of the stator of the skipping motor.

Skipping motor 800 optionally comprises a stator 840 having four electromagnets 841A, 841B, 841C and 841D only two of which, electromagnets 841A and 841C, that are opposite each other are shown in the figure. Electromagnets 841A, 841B, 841C and 841D are optionally arranged in a rotationally symmetric configuration having an angle of rotational symmetry equal to $2\pi/4$. The skipping motor comprises a rotor 830 mounted to a stem 850 having a ball 854 that seats in a matching socket 832 formed in the rotor in accordance with an embodiment of the disclosure. Dimensions and tolerances of ball 854 and socket 832 are determined to allow nutating skipping rotation of rotor 830. Optionally, rotor 830 rotatably rests on at least three bearing posts that may be raised and lowered to control attitude of rotor 830 and an angle between an axis 833 of rotation of the rotor and axis 845 of stator 840. Skipping motor 800 is assumed by way of example to comprise four bearing posts 861A, 861B, 861C, and 861D of which in FIGS. 9A and 9B only bearing posts 861A, and 861C are shown. Rotor 830 has an edge surface 834 that is a spherical zone having center C located at a center of ball 854 and a radius R_K . Electromagnets 841A, 841B, 841C, and 841D have stator contact surfaces 847A, 847B, 847C, and 847D respectively, each of which is a portion of a surface of sphere having a radius R_S and center C so that the stator contact surfaces are concentric with spherical zone edge surface 834 of rotor 830. In FIGS. 9A and 9B only stator contact surfaces 847A and 847C are shown.

Because edge surface 834 and stator contact surfaces 847A, 847B, 847C, and 847D are concentric, attitude of axis of rotation 833 of rotor 830 relative to axis 845 may be changed without changing the skipping ratio and thereby skip angle of skipping motor 800. FIG. 9A schematically shows skipping motor 800 for which bearing posts support rotor 830 with rotor axis 833 substantially parallel to axis 845 of stator 840. FIG. 9B schematically shows skipping motor 800 for which bearing posts support rotor 830 with rotor axis 833 at an angle θ relative to axis 845.

It is noted that in the above discussion skipping motors are described as having N electromagnets and contact surfaces that are rotationally symmetric and having an angle of rotational symmetry and angle between locations of the electromagnets equal to $2\pi/N$. However, practice of an embodiment of the disclosure is not limited to rotational symmetries for which electromagnets are angularly equally spaced. For example, an angle between a first pair of angularly adjacent electromagnets of a skipping motor in accordance with an embodiment of the disclosure may be different than an angle between a second pair of electromagnets in the skipping motor. As a result of the difference, a skip angle between the first pair will be different from the skip angle between the second pair.

It is further noted that in the above discussion skipping motors are described as having a rotor having cylindrical surface and having N electromagnets with contact surfaces

that are cylindrical in shape. In an embodiment, the rotor may have a surface that is shaped as a screw, and the contact surfaces of the electromagnets may have contact areas made in a shape of a part of a nut. Skipping the motor will create axial movement of the rotor as well as rotation.

In the description and claims of the present application, each of the verbs, "comprise" "include" and "have", and conjugates thereof, are used to indicate that the object or objects of the verb are not necessarily a complete listing of components, elements or parts of the subject or subjects of the verb.

Descriptions of embodiments of the invention in the present application are provided by way of example and are not intended to limit the scope of the invention. The described embodiments comprise different features, not all of which are required in all embodiments of the invention. Some embodiments utilize only some of the features or possible combinations of the features. Variations of embodiments of the invention that are described, and embodiments of the invention comprising different combinations of features noted in the described embodiments, will occur to persons of the art. The scope of the invention is limited only by the claims.

The invention claimed is:

1. A skipper motor comprising:

- a stator and a rotor;
- a plurality of at least three electromagnets mounted to one of the stator or the rotor, each of the plurality of the at least three electromagnets comprising a ferromagnetic core having at least one contact surface, wherein the stator or rotor has an axis of rotational symmetry about which axis of rotational symmetry the stator or rotor is rotatable so that the at least one contact surface of any of the plurality of ferromagnetic cores rotates to substantially congruently locate to at least a portion of a location of the at least one contact surface of any other of the plurality of ferromagnetic cores;
- a coupling of the rotor to the stator configured to enable rotation of the rotor relative to the stator and contact of the stator and rotor along a contact surface of the at least one contact surface of an electromagnet of the plurality of at least three electromagnets during operation of the motor; and
- a controller operable to energize electromagnets of the plurality of electromagnets to cause the rotor to rotate from a first to a second motor state of the skipper motor;

wherein, in the first motor state the stator and rotor contact along a first pair of contact surfaces comprising a contact surface of the at least one contact surface from each ferromagnetic core of first and second rotationally adjacent electromagnets of the plurality of at least three electromagnets mounted to the stator or rotor and matching surface regions of the rotor or stator respectively, and in the second motor state the stator and rotor are in contact along a second pair of contact surfaces comprising a contact surface of the at least one contact surface from each ferromagnetic core of the first electromagnet and a third electromagnet of the plurality of at least three electromagnets that is rotationally adjacent the first electromagnet and matching surface regions of the rotor or stator respectively; and

to initiate rotation from the first to the second motor state the controller simultaneously energizes the first and third electromagnets with respective first and third currents so that magnetic flux generated in the ferromagnetic core of the third electromagnet substantially

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cancels magnetic flux generated in the ferromagnetic core of the second electromagnet by energization of the third electromagnet.

2. The skipper motor according to claim 1 wherein energizing the first and third electromagnets to substantially cancel magnetic flux in the second electromagnet comprises energizing the first and third electromagnets so that a current I_1 flowing in a coil of the first electromagnet and a current I_2 flowing in a coil of the third electromagnet satisfy a relationship:

$$I_1/I_2 = -1/(1 + \delta \mu(m)/L_O),$$

wherein the stator and rotor are formed from ferromagnetic materials having a same permeability $\mu(m)$, as that of the ferromagnetic cores and in the relationship L_O is a length of a magnetic circuit through the ferromagnetic cores of the first and second electromagnets and ferromagnetic material of the rotor and stator when there is substantially no air gap in the circuit, δ is a length of an airgap in a magnetic circuit through the ferromagnetic core of the third electromagnet when there is no air gap in the magnetic circuit through the ferromagnetic cores of the first and second electromagnets.

3. The skipper motor according to claim 1 wherein after initiation the contact surface of the second electromagnet separates from contact with the rotor or stator and the controller operates to energize the first and third electromagnets with currents having substantially equal magnitude.

4. A skipper motor according to claim 1:

wherein the number of the plurality of at least three electromagnets is equal to three, the stator and rotor are formed from ferromagnetic materials having a same permeability $\mu(m)$, as that of the ferromagnetic cores, and when there is substantially no air gap in a magnetic circuit through the ferromagnetic cores of the first and second electromagnets, the magnetic circuit has a length L_O , a magnetic circuit through the third electromagnet has an air gap of length δ , and L_O and δ satisfy a relationship $\delta < 3L_O \mu(m)$.

5. The skipper motor according to claim 1 wherein the ferromagnetic cores have magnetic coercivity less than or equal to about 10 Oersted (Oe), about 1 Oe, or about 0.5 Oe.

6. The skipper motor according to claim 1 wherein the at least one contact surface of the ferromagnetic core of each electromagnet is formed having two contact surfaces.

7. The skipper motor according to claim 6 wherein each of the two contact surfaces is a curved surface having a same radius of curvature.

8. The skipper motor according to claim 7 wherein a first one and a second one of the two contact surfaces of the two contact surfaces have different centers of curvature.

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9. The skipper motor according to claim 1 wherein the at least one contact surfaces of the plurality of at least three electromagnets are configured in an array that exhibits an N-fold symmetry of rotation and a symmetry angle of rotation equal to $2\pi/N$, where N is the number of the plurality.

10. The skipper motor according to claim 1 wherein the plurality of at least three electromagnets are mounted to the stator or rotor and the ferromagnetic cores are formed having shelves that support the rotor or stator respectively and comprising a spacer made from a non-ferromagnetic and relatively low friction material between the rotor or stator and each shelf that supports the stator or rotor respectively.

11. The skipper motor according to claim 1 wherein perpendicular to the axis of rotation the radius of curvature of the at least one contact surfaces of the ferromagnetic cores about the axis of rotation is a function of location along the axis of rotation.

12. The skipper motor according to claim 11 wherein the rotor or stator is moveable to position the rotor at different locations along the axis of rotation.

13. The skipper motor according to claim 1 wherein the at least one contact surfaces of the ferromagnetic cores of the plurality of at least three electromagnets are surface portions of a same first spherical surface, the matching surfaces are surface portions of a same second spherical surface and the first and second spherical surfaces have substantially a same center.

14. The skipper motor according to claim 13 wherein the stator and rotor are rotatable relative to each other about the center of the spherical surfaces.

15. The skipper motor according to claim 1 wherein the at least one contact surfaces of the plurality of at least three electromagnets lie along a first protruding or recessed helix having a pitch and the matching surfaces lie along a second recessed or protruding helix respectively that is meshed with the first helix and has a same pitch as the first helix.

16. A skipper motor according to claim 1 wherein all the at least one contact surfaces of the plurality of at least three electromagnets lie on a same circularly cylindrical or a spherical surface.

17. The skipper motor according to claim 16 wherein spaces between the ferromagnetic cores of the plurality of at least three electromagnets are filled with a filler material.

18. The skipper motor according to claim 17 wherein the filler material has surface regions that span distances between the at least one contact surfaces and lie on the same surface on which same surface all the contact surfaces lie.

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